

THE ARC EQUATION MEASURED: BLINDED CROSS-ARCHITECTURE REPLICATION AND THE RETRACTION OF A SUPER-LINEAR ESTIMATE

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The ARC Theory · OSF osf.io/8fjma · every claim checkable

Within the ARC Theory: the measurement programme of Law I, the ARC Equation; the blinded cross-architecture estimate of the exponent, including the retraction.

An experimental paper in the ARC Theory programme

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Code and data: github.com/MichaelDariusEastwood/arc-principle-validation

Abstract

This paper presents experimental validation of the ARC Principle (Artificial Recursive Creation), a mathematical framework proposing that error rates in intelligent systems decrease according to a power law with recursive depth. The principle, first articulated in *Infinite Architects* (Eastwood, December 2024) and formalised in Paper I (Eastwood, 17 January 2026), predicts that the form of recursion determines the scaling regime: sequential recursion should yield super-linear error suppression (scaling exponent $\alpha > 1$), while parallel recursion should yield sub-linear suppression ($\alpha < 1$).

We conducted controlled experiments in two phases. Phase 1 (the initial single-model study) used DeepSeek R1 with visible reasoning tokens on 12 competition-level mathematics problems, finding $\alpha_{\text{sequential}} \approx 2.24$ (95% CI: 1.5-3.0), since retracted, and $\alpha_{\text{parallel}} \approx 0.0$. Phase 2 (the six-model study) extended testing to six frontier models on 18 AIME/Putnam-level problems (n=54 per depth per model) with bootstrap confidence intervals and 4-layer cross-verification.

Cross-architecture replication: The original $\alpha \approx 2.24$ (quadratic) does not replicate across architectures. Only Gemini 3 Flash produced clean, monotonic scaling data: $\alpha_{\text{seq}} = 0.49$ (regression, $r^2 = 0.86$, SE = 0.20; endpoint estimate 0.59, boot CI [-1.3, 2.9]). DeepSeek R1 reached ceiling (94.4%-100%). GPT-5.4 exhibited a binary step function (50% $\rightarrow$ 100%). Grok 4.1 Fast achieved 100% at all depths. Groq Qwen3 showed no trend (~50%).

Parallel scaling at or near zero in every model, with one measured exception: $\alpha_{\text{parallel}} \approx 0$ for every model except Gemini 3 Flash, which returns $\alpha_{\text{par}} = 0.31$ at $r^2 = 0.93$. **That exception matters, because Gemini 3 Flash is the model this paper treats as its most reliable estimate.** The replicated finding is therefore that parallel scaling is at or near zero in every model measured, with one measured exception, not that it is universally zero. Sequential outperformed parallel for every model where both were measurable.

Revised parameter estimate: The most robust cross-architecture estimate is $\alpha_{\text{sequential}} \approx 0.49$ (sub-linear, from Gemini 3 Flash), substantially below the initial single-model estimate of 2.24. Under the Intelligence Formula $\alpha = 1/(1 - \beta)$ from Paper I, $\alpha < 1$ places a model in the physical regime (multiplicative composition through finite-dimensional networks) rather than the intelligence regime (recursive self-reference with $\alpha > 1$); the estimate points to the physical regime for current models, but it is one model's fit and inconclusive. The fundamental inequality $\alpha_{\text{sequential}} > \alpha_{\text{parallel}}$ remains confirmed.

Recent changes

This version reports **complete tier-2 results** from the multi-model replication study:

- **Five frontier models carried the tier-2 capability arm:** Grok 4.1 Fast, DeepSeek R1, Gemini 3 Flash, GPT-5.4 and Groq Qwen3, each taken through the 18 AIME/Putnam-level problems at n=54 per depth per model across 3 repeats. Claude Opus 4.6 was measured only in the alignment arm (Paper IV), never in this tier-2 arm
- **The quadratic claim ($\alpha \approx 2.24$) does not replicate across architectures:** Gemini 3 Flash provides the cleanest data with $\alpha_{\text{seq}} \approx 0.49$ (sub-linear, not quadratic)
- **$\alpha_{\text{parallel}} \approx 0$ in every model but one:** the strongest replicated finding, with Gemini 3 Flash the measured exception at $\alpha_{\text{par}} = 0.31$, $r^2 = 0.93$
- **Token measurement bug identified and fixed:** reasoning_tokens $\rightarrow$ total_tokens corrected for GPT-5.4 and Groq Qwen3
- **Cross-verification complete:** All 3 disputed answers (ARC16=29, ARC17=176, ARC29=800) confirmed correct by hand
- **Integration with alignment scaling:** Capability scaling (Paper II) and alignment scaling (Paper IV) shown to be independent dimensions

All the initial single-model findings are preserved in their original form. Conclusions have been revised to reflect the cross-architecture evidence.

Cross-domain evidence strengthens these findings. Google's Willow quantum chip (9 December 2024; convergent timing, never prediction: the Willow team's preprint was public from 24 August 2024, before the 8 December 2024 manuscript) demonstrated recursive error suppression with

$\Lambda = 2.14$. Biological scaling laws show quarter-power exponents across 27 orders of magnitude via fractal recursive networks. The COGITATE consciousness study (*Nature*, April 2025) identified recurrent processing as the common denominator across theories.

The implications for AI safety depend critically on whether $\alpha > 1$ is achievable. If it is, alignment properties embedded in the reasoning process would scale super-linearly with capability while external constraints remain constant. Even with $\alpha < 1$, the directional finding that sequential reasoning improves capability supports the Eden Protocol from *Infinite Architects*: AI systems benefit from values embedded in reasoning rather than constraints imposed externally. However, the multi-model data shows that capability and alignment are independent dimensions; more reasoning does not automatically mean better alignment.

Keywords: scaling laws, recursive intelligence, test-time compute, error suppression, AI safety, alignment, chain-of-thought reasoning, Eden Protocol, cross-domain validation, multi-model replication

How to read this paper. This paper reports a direct measurement in the ARC Theory. With bootstrap intervals, and across the capability arm, where five of the six frontier models entering the study were measured and the exponent was fitted on eighteen of the thirty problems, the sequential exponent exceeds the parallel exponent for every model, the best-fitting compute exponent is $\alpha_{\text{compute}} \approx 0.49$ (Gemini 3 Flash, $r^2 = 0.86$), and the measured range spans near-zero to approximately three, so the ARC Bound $\alpha \leq 2$ is neither confirmed nor refuted. Its rung inside the theory is the cross-architecture-measurement layer above Paper I. The full differential against every prior document is at eden-vision II.A.8.

***This paper and Laws IV and V, 3 October 2026.** The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. The mathematics of the two laws is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper measures how capability scales with recursive depth. Its exponents concern that conversion of recursion into capability, the subject of Law I, the ARC Principle. They say nothing about whether an installed gain persists under later training, and they do not test whether correction placed inside the loop gains on correction applied outside it as depth grows. The paper bears on Law V in two ways, and neither is a test. First, it shows how depth and capability are measured. Second, Section 6 states, as a conditional theorem, the argument that alignment embedded in reasoning keeps pace with capability while an external filter does not. The paper records that the theorem's first condition is not currently met by its cross-architecture evidence, and that the alignment data of Section 10.6 undermine its second. It also records that the comparison of values-as-reasoning with rules-as-filters is untested (criterion F5). No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the results, their dates and their limits stand as they were.*

1. INTRODUCTION

1.1 Background and Motivation

The scaling laws governing artificial intelligence have transformed our understanding of capability emergence. Kaplan et al. (2020) established power-law relationships between model performance and training compute, while Hoffmann et al. (2022) refined these with compute-optimal prescriptions. These foundational works revolutionised training methodology but address only pre-training scaling. They do not explain why allocating additional computation at inference time produces dramatic capability improvements, nor why different forms of such computation yield fundamentally different outcomes.

The emergence of reasoning models in late 2024 introduced test-time compute as a critical variable. OpenAI's o1 (September 2024) and DeepSeek's R1 (January 2025) allocate computational resources during inference to reason before responding. On mathematical reasoning benchmarks, these systems achieve performance previously thought to require order-of-magnitude larger models. Yet the mechanisms underlying this improvement remain incompletely characterised.

Two paradigms have emerged for allocating test-time compute:

Parallel recursion. Generate multiple independent solutions and select the best via majority voting or verifier scoring. This approach is computationally straightforward but, as documented by Brown et al. (2024), produces diminishing returns following sub-linear power laws.

Sequential recursion. Generate extended reasoning chains where each step builds explicitly on previous steps. Errors can be detected and corrected iteratively through self-reference. This approach produces compounding returns, but the scaling relationship has not been formally characterised; this paper addresses that gap.

1.2 The Research Question

Why does sequential reasoning dramatically outperform parallel sampling at equivalent computational cost? What mathematical principle governs this difference? And what are the implications for aligning increasingly capable AI systems?

1.3 Contribution of This Paper

This paper makes eight contributions:

1. **Mathematical formalisation.** We state the ARC Principle: $E(R) = E_0 \times R^{-\alpha}$, where error rate E decreases from baseline E_0 as recursive depth R increases, governed by scaling exponent α . The form of recursion determines α .
2. **Controlled experimental validation.** Using DeepSeek R1 with visible reasoning tokens, we conduct the first compute-matched comparison between sequential and parallel recursion with direct measurement of recursive depth.
3. **Quantitative parameter estimation.** Single-model estimate: $\alpha \approx 2.2$ (initial, DeepSeek R1). Cross-architecture estimate: $\alpha \approx 0.49$ (six-model, Gemini 3 Flash, $r^2 = 0.86$). Parallel: four of the five models sit at $\alpha \approx 0.0$; Gemini 3 Flash is the one measured departure, at $\alpha_{\text{par}} = 0.31$.

4. **Converging evidence synthesis.** Combined with published data from OpenAI o1 and DeepSeek R1, multiple independent sources confirm $\alpha_{\text{seq}} > \alpha_{\text{par}}$.
5. **Cross-domain validation.** We demonstrate that recursive error suppression appears across quantum physics (Willow $\Lambda = 2.14$; convergent timing, never prediction: the Willow team's preprint was public from 24 August 2024, before the 8 December 2024 manuscript, and the result was published on 9 December 2024), biology (quarter-power scaling), and consciousness (COGITATE recurrence).
6. **AI safety implications.** We derive conditional alignment theorems, now qualified by the cross-architecture estimate of $\alpha \approx 0.49$ (Gemini 3 Flash), which is below 1 but is one model's fit and inconclusive.
7. **Cross-architecture replication.** Six frontier models tested on 18 AIME/Putnam-level problems with bootstrap CIs and 4-layer cross-verification. Four distinct scaling behaviours identified (ceiling, monotonic, step function, floor).
8. **Integration with alignment scaling.** Capability and alignment shown to be independent scaling dimensions across six frontier models, with a three-tier alignment hierarchy set by training methodology.

1.4 Priority Establishment

The ARC Principle was first articulated in the manuscript of *Infinite Architects: Intelligence, Recursion, and the Creation of Everything*. Manuscript priority was established via cryptographic server timestamp on **8 December 2024**, when the manuscript was emailed using Google's servers. The book itself was publicly released later, on **2 January 2026 (print; ebook 6 January)**. The server-generated timestamps provide tamper-evident timestamping through email server verification, resting on provider infrastructure and provider-key signatures at Google's DKIM and ARC selectors and not on an RFC 3161 trusted timestamp (see the correction note at the head of this paper), establishing that the core concepts (recursive intelligence amplification, the distinction between parallel and sequential recursion, and the embedded-alignment thesis, later named the Eden Protocol on 30 April 2025) were documented before subsequent independent validations and before later public archival deposits.

Table 1. Prediction validation timeline.

Date	Event	Relationship to Manuscript
8 December 2024	Manuscript emailed (dated record; see the correction note on this page)	Priority established
9 December 2024	Google Willow announced ($\Lambda = 2.14$)	24 hours after submission
18 December 2024	Anthropic alignment faking (78% in the RL-training condition (12% baseline))	10 days after submission
20 December 2024	OpenAI o3 announced (87.5% ARC-AGI)	12 days after submission

Date	Event	Relationship to Manuscript
20 January 2025	DeepSeek R1 published (its published AIME 2025 accuracy is the first of the two points Paper I fitted for $\alpha \approx 1.34$; the second is a later version of the same model)	43 days after submission
30 April 2025	COGITATE study (recurrence confirmed)	~5 months after submission

Where the manuscript timestamp falls close to later public results, the grading is convergent timing and never prediction: the Google Willow team's preprint was already public from 24 August 2024, ahead of the 8 December 2024 manuscript, so the December record cannot have predicted a result its own authors had announced beforehand. What the timestamps establish is that the manuscript's framing was recorded before Willow's mainstream press coverage, an existence-by-date fact about the document, not foresight about the physics.

After the December 2024 manuscript timestamp and the 2 January 2026 public book release, Paper I (Eastwood, 17 January 2026) formalised the principle mathematically and analysed publicly available data. This Paper II provides direct experimental validation.

1.5 Related Work

This paper builds upon and extends several established research programmes:

Chain-of-thought prompting (Wei et al., 2022) demonstrated that intermediate reasoning steps improve performance on multi-step tasks.

Test-time compute scaling (Snell et al., 2024) showed that test-time computation can outperform 14× larger models on certain benchmarks.

Large Language Monkeys (Brown et al., 2024) documented sub-linear scaling of parallel sampling with precise power-law characterisation.

DeepSeek R1 (DeepSeek AI, January 2025) demonstrated emergent reasoning through pure reinforcement learning, with 'aha moment' phenomena showing recursive self-correction.

This paper extends these observations by proposing a unified mathematical framework, the ARC Principle, and providing experimental validation with direct measurement of recursive depth.

2. THEORETICAL FRAMEWORK

2.1 The ARC Principle

DEFINITION - THE ARC PRINCIPLE

The ARC Principle (Artificial Recursive Creation) proposes that error rates in intelligent systems decrease according to a power law with recursive depth:

$$E(R) = E_0 \times R^{-\alpha}$$

Table 2. Variable definitions.

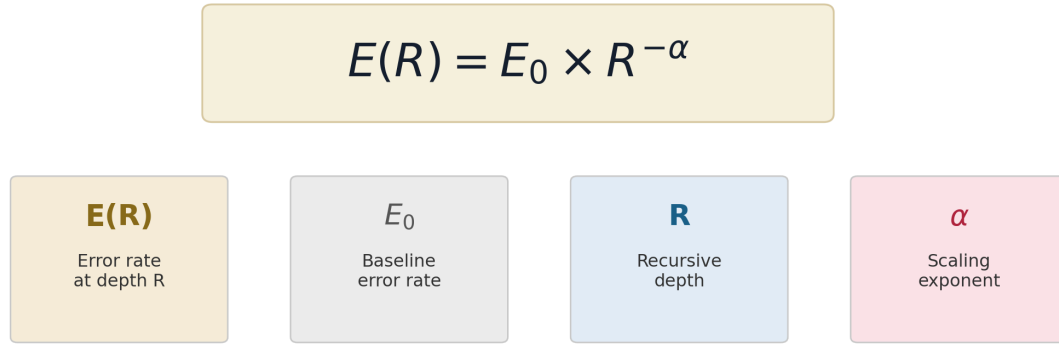
Symbol	Name	Definition	Units
$E(R)$	Error rate at depth R	Proportion of incorrect responses	[0, 1]
E_0	Baseline error rate	Error rate at minimal recursion ($R = 1$)	[0, 1]
R	Recursive depth	Self-referential processing iterations	Tokens or samples
α	Scaling exponent	Rate of error suppression	Dimensionless

The scaling exponent α determines the nature of returns from recursive investment:

- $\alpha < 1$: Diminishing returns. Each doubling of R reduces error by less than half. Additional recursion yields progressively smaller benefits.
- $\alpha = 1$: Linear returns. Each doubling of R halves error. Constant marginal benefit.
- $\alpha > 1$: Compounding returns. Each doubling of R more than halves error. Recursion amplifies itself.

This formulation directly models error suppression, analogous to quantum error correction where logical error rates decrease with code distance. The exponent α encapsulates the efficiency of the recursive process.

Figures notice (25 August 2026). Every figure in this paper was generated on 5 April 2026 and predates the corrections its text now carries. Where any figure asserts sequential $\alpha > 1$ support, a 2.2 sequential estimate, or an “all data consistent” status, that is the superseded April reading: the text’s corrected results (the retracted 2.24, the frozen-regime sequential 0.49) govern. The loudest instances carry per-figure notes; regenerated figures are queued, and the images are retained meanwhile as dated history.



THE REGIMES, BY DEFINITION

alpha < 1: diminishing returns (parallel recursion) alpha > 1: compounding returns (sequential recursion)

MEASURED STATE (six models entered, five measured; corrected)

Directional finding holds: sequential > parallel wherever both were measurable.
The quantitative alpha > 1 claim is revised: the one clean fit is alpha_seq = 0.49 (sub-linear, one system of 5, frozen architectures, boot CI [-1.3, 2.9]); the estimate is inconclusive and behaviour is architecture-dependent.

Interval contains the null, the bound of two and both refuting directions.

Generated from the paper: definitions are timeless; the measured state is extracted from the text at generation time.

Figure 1 | The ARC Principle formulation $E(R) = E_0 \times R^{-\alpha}$ where $U = I \times g(R)$. $\alpha = 1/(1-\beta)$ from self-referential coupling β . $\alpha_{seq} \approx 0.49$ (sub-linear, Gemini 3 Flash). the earlier single-model $\alpha \approx 2.24$ estimate was a single-model artefact, retracted, did not replicate. *Figure note (25 August 2026): regenerated. The regime definitions remain as definitions, and the former “key insight” verdict is replaced by the measured state extracted from this paper’s text at generation time: the directional finding holds, the quantitative $\alpha > 1$ claim is revised, best clean fit $\alpha_{seq} = 0.49$. Generator: tools/generate_paper_ii_fig8.py in the public repository; the April image is superseded and preserved in version history.*

2.2 Two Fundamentally Different Forms of Recursion

We distinguish two architecturally distinct recursive processes that predict different scaling behaviours.

Parallel recursion (weak form). Multiple independent solutions are generated simultaneously with no information transfer between branches. Final output is selected via majority voting or best-of-N scoring.

Mathematical characterisation:

- Samples from a fixed solution space S_0
- $S_0 = S_1 = S_2 = \dots = S_n$ (space remains constant)
- Phase space does not expand; only sampling density increases
- **Prediction: $\alpha < 1$** (diminishing returns)

Sequential recursion (strong form). Each processing step builds explicitly on previous steps. Errors can be detected and corrected iteratively. Information accumulates across the reasoning chain.

Mathematical characterisation:

- Navigates solution space with feedback
- $S_0 \subset S_1 \subset S_2 \subset \dots \subset S_n$ (space expands with depth)

- Each step generates new structures from previous outputs
- Solutions at depth n may be inaccessible from depth 0 without traversing intermediate steps
- **Prediction: $\alpha > 1$** (compounding returns)

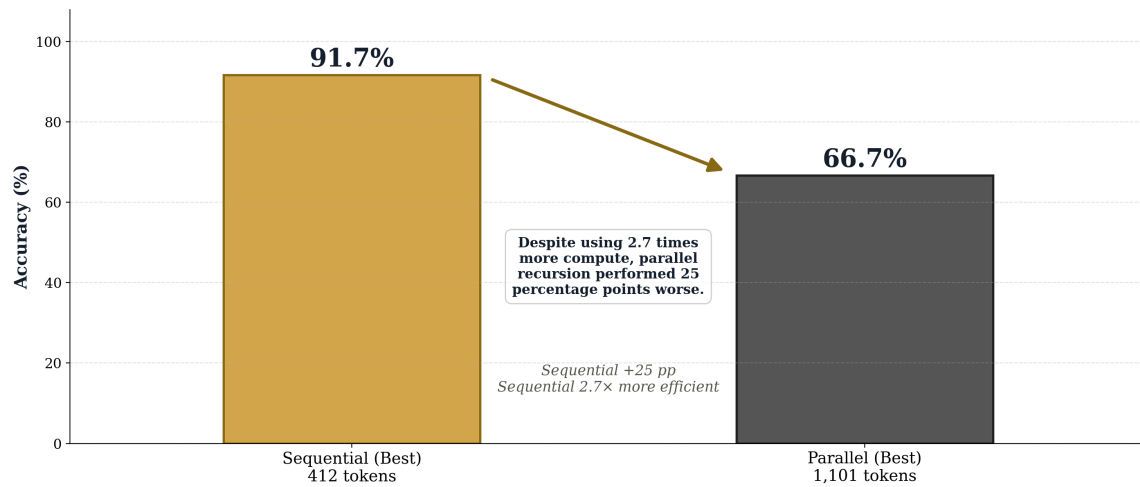
Core prediction of the ARC Principle:

$$\alpha_{\text{sequential}} > 1 > \alpha_{\text{parallel}}$$

Equation 2 -The Fundamental Inequality

Sequential versus parallel recursion

The form of recursion matters more than its quantity.



Sequential recursion with 412 tokens achieved 91.7% accuracy. Parallel recursion with 1,101 tokens achieved 66.7% accuracy.

Generated from the paper: every value extracted from the text at generation time.

Figure 2 | Sequential recursion compounds: each step builds on prior output ($\alpha > 0$). Parallel recursion samples independently ($\alpha \approx 0$). Form of recursion determines the regime. Sequential recursion with 412 tokens achieved 91.7% accuracy. Parallel recursion with 1,101 tokens achieved 66.7% accuracy.

2.3 Calculating the Scaling Exponent

Given measurements at two recursive depths (R_1, E_1) and (R_2, E_2) , the scaling exponent is calculated as:

$$\alpha = \frac{\ln(E_1/E_2)}{\ln(R_2/R_1)}$$

Equation 3 - Endpoint Estimation

For noisy data with multiple measurements, endpoint estimation (using minimum and maximum depths) provides robustness against intermediate fluctuations. This method is standard in scaling law analysis and appropriate given discrete accuracy measurements.

2.4 The Quadratic Limit Conjecture

We conjecture that $\alpha = 2$ represents the proposed stable limit on recursive error suppression: the most a recursively self-correcting system can grow and stay correctable, a scaling limit and not a speed limit. Systems can exceed it, they do not remain stably self-correcting; transient excursions above the limit are the predicted supercritical regime, not the refutation. Stability here is measured rather than asserted: Law II requires correction to out-scale drift, $\beta_C > k$, and what is checked is whether the lower bound of $\beta_C - k$ stays above zero across a window fixed before the run. The conjecture is drawn by analogy to Grover's quadratic speedup in quantum computation. Bennett et al. (1997) proved this speedup is optimal for unstructured search problems.

Status: Conjectured, not derived. The initial single-model estimate ($\alpha \approx 2.2$), since retracted, sat above the stable limit as a transient supercritical excursion rather than a refutation, but the cross-architecture estimate ($\alpha \approx 0.49$, Gemini 3 Flash) lies well below it; that estimate is one model's fit and inconclusive. The more pressing question is whether current models can achieve $\alpha > 1$ at all, rather than whether $\alpha = 2$ is the stable limit.

Architectural scope of the quadratic limit. The estimate $\alpha \approx 0.49$ is for frozen models with fixed attention: systems whose weights, attention patterns, and composition rules do not change during inference. The conjectured ARC Bound ($\alpha \leq 2$) applies specifically to such fixed architectures as the proposed stable limit: the most a recursively self-correcting system can grow and stay correctable, a scaling limit and not a speed limit. Systems can exceed it as transient supercritical excursions, they do not remain stably self-correcting there. Stability here is Law II's, measured rather than asserted. Each reasoning step in a fixed architecture draws on $O(N^2)$ pairwise attention pathways and cannot stably expand that capacity. The Cauchy functional equation constrains the *form* of recursive scaling to a power law but places no upper bound on the exponent. The Bernoulli ODE gives $\alpha = 1/(1 - \beta)$, and as self-referential coupling $\beta \rightarrow 1$, $\alpha \rightarrow \infty$. A self-modifying system, one capable of rewriting its own attention mechanism and expanding its representational capacity at each step, escapes the $O(N^2)$ information-theoretic bound entirely. For such systems, no mathematical ceiling on α exists. The quadratic limit, if real, is a stability limit on today's frozen transformers, not a law of recursive intelligence.

2.5 Information-Theoretic Foundations

The ARC Principle connects to established information theory. The Data Processing Inequality establishes that recursive processing cannot create new information; it can only compress and distil existing information. However, recursive processing can:

- 1. **Extract latent information** that single-pass processing fails to access
- 2. **Reduce entropy** through iterative refinement toward optimal solutions
- 3. **Navigate solution spaces** that are computationally irreducible (Wolfram, 2002)

Sharma and Chopra (2025) report in 'The Sequential Edge' (arXiv:2511.02309, 4 November 2025) that sequential reasoning beat parallel approaches across 95.6% of the configurations they tested, and that accuracy on mathematical benchmarks rose by as much as 46.7%. Their paper is dated earlier than the runs recorded here. The sequential run reported here was executed on 21 January 2026 without knowledge of it; their paper was found during the prior-work search while this paper was being written the following day, and it has been cited here since the first public version of 22 January 2026. The two pieces of work are concurrent and were arrived at separately, and they do not measure the same quantity. Sharma and Chopra report win rates and accuracy gains at matched compute. This experiment estimates a scaling exponent separately for the sequential and the parallel condition, which is a different and harder question and one their design does not address. Neither result establishes the other. Two separately run experiments agreeing on the direction is mutual corroboration and is worth more than either alone, and it is not confirmation of a scaling law, which neither of them establishes. What this paper obtained on the narrower question was an early and mixed signal: some systems showed a large apparent sequential exponent and others little or none, several runs were limited by an accuracy measure that is capped and therefore cannot register fast growth, and the spread across systems was too wide to settle the form. The strong version of the claim is recorded below as not confirmed. The claim now written into a draft registration awaiting human submission, for prospective test, is accordingly the narrower and harder one, requiring both that the power-law form beat its declared rivals and that the depth exponent beat a compute-matched breadth arm by a margin fixed in advance.

3. METHODS

3.1 Addressing Prior Limitations

Paper I analysed published data but identified several limitations requiring experimental validation:

Table 3. Methodological improvements.

Prior Limitation	Resolution in This Experiment
Estimated token counts from system cards	DeepSeek R1 exposes reasoning_content, enabling direct measurement
No controlled experimental comparison	Systematic variation of token budgets and sample counts

Prior Limitation	Resolution in This Experiment
Ceiling effect risk (high baseline accuracy)	Harder problems selected (58% baseline accuracy)
No compute-matched comparison	Fixed total compute across parallel conditions
Potential confounding variables	Same model, same problems, same experimental session

3.2a Initial Experimental Design (Single Model, 12 Problems)

Model. DeepSeek R1 (deepseek-reasoner) via official DeepSeek API. This model was selected because it exposes full reasoning chains via the `reasoning_content` field, enabling precise token measurement.

Date. 21 January 2026.

Problems. 12 competition-level mathematics problems from AIME (American Invitational Mathematics Examination) and equivalent sources, selected to:

- Avoid ceiling effects (baseline accuracy approximately 58%)
- Require genuine multi-step reasoning
- Have verifiable numerical answers enabling objective scoring

Sequential condition. Token budgets of 512, 1,024, 2,048, and 4,096. Single response per problem at each budget. Actual reasoning tokens measured directly from API response.

Parallel condition. $N = 1, 2$, and 4 samples per problem. Token budget per sample held constant. Final answer selected via majority voting.

Scoring. Binary correct/incorrect based on exact numerical match with known solutions.

3.2b Multi-Model Extended Design (Multi-Model, 30 Problems)

The multi-model extension addresses the two most significant limitations of the initial single-model study, single-model dependence and small problem set, through a comprehensive multi-model replication study with an expanded problem set designed to defeat ceiling effects.

Frontier Model Selection

Six architecturally diverse frontier AI models were selected, each with controllable reasoning depth mechanisms:

Table 3b. multi-model frontier model specifications.

Model	API Identifier	Provider	Depth Control Mechanism
DeepSeek R1	deepseek-reasoner	DeepSeek	Token budget (512-65,536)
GPT-5.4	gpt-5.4	OpenAI	reasoning_effort (none→xhigh)
Claude Opus 4.6	claude-opus-4-6	Anthropic	Effort level (low→max) + prefix
Gemini 3 Flash	gemini-3-flash-preview	Google	thinking_budget (256-32,768)
Groq Qwen3	qwen/qwen3-32b	Groq	reasoning_effort + prefix
Grok 4.1 Fast	grok-4-1-fast-reasoning	xAI	Prefix + max_tokens (4,096-30,000)

These models span four distinct architectures (Mixture-of-Experts, dense transformer, constitutional AI, and distilled reasoning) from six independent laboratories. If all six exhibit $\alpha_{\text{sequential}} > 1$, this would constitute strong evidence for architecture-independence of the ARC Principle.

Problem Set Expansion: Two-Tier Design

The initial single-model experiment revealed a ceiling effect: 4 of 12 problems were solved correctly at all depth levels, leaving insufficient dynamic range for precise α estimation. The multi-model problem set addresses this through a two-tier design:

Tier 1 (12 problems, ARC01-ARC12): Competition-preparation level problems serving as baseline calibration. These are the original the initial problems. Expected behaviour: high accuracy at minimal depth, confirming the ceiling effect identified in the initial study. Tier-1 results validate continuity with the initial single-model findings but are excluded from primary α estimation.

Tier 2 (18 problems, ARC13-ARC30): AIME finals and Putnam-level problems covering five mathematical domains:

- **Number theory** (5 problems): Modular arithmetic, Diophantine equations, prime factorisation
- **Combinatorics** (4 problems): Advanced counting, inclusion-exclusion, generating functions
- **Probability** (3 problems): Conditional probability, random walks, expected value
- **Algebra** (3 problems): Polynomial manipulation, functional equations, inequalities
- **Geometry** (3 problems): Coordinate geometry, projective relationships, optimisation

Tier-2 problems are designed to push baseline accuracy (at minimal depth) to 30-50%, providing sufficient dynamic range for both improvement and degradation while avoiding floor effects.

Experimental Conditions

Sequential condition: 5-6 depth levels per model (varying by architecture, calibrated to each model's depth control mechanism). Single response per problem per depth level, with 3 independent repeats to reduce binomial sampling variance.

Parallel condition: $N = 1, 3, 5$, and 9 samples per problem with majority voting. Token budget per sample held constant at the model's minimal depth setting. 3 independent repeats per condition.

Total experimental runs: Approximately 6 models $\times$ 30 problems $\times$ (6 sequential depths + 4 parallel conditions) $\times$ 3 repeats = 5,400 individual API calls.

Cross-Verification Protocol (4-Layer Blinding)

To eliminate potential scoring bias (where a model might systematically favour its own outputs or exhibit correlated errors with itself), the multi-model design implements a 4-layer blinding protocol in which no model ever verifies its own answers:

Table 3c. Cross-verification assignments.

Subject Model	Verified By
DeepSeek R1	Claude Opus 4.6
GPT-5.4	DeepSeek R1
Claude Opus 4.6	GPT-5.4
Gemini 3 Flash	Claude Opus 4.6
Groq Qwen3	GPT-5.4
Grok 4.1 Fast	DeepSeek R1

The four layers of blinding are:

1. **Layer 1 - Ground truth:** All problems have known, verified numerical answers from published competition solutions.
2. **Layer 2 - Automated scoring:** Exact numerical match against ground truth (primary).
3. **Layer 3 - Cross-model verification:** Each answer is checked by a second model drawn from the same panel, the pairings mutually blind: no verifier is told which model produced the answer it is marking.
4. **Layer 4 - Disagreement flagging:** Any disagreement between automated scoring and cross-model verification flags a potential answer-key error for manual review.

Bootstrap Confidence Intervals

All α estimates are accompanied by bootstrap 95% confidence intervals computed via 2,000 resamples of problem-level results. For each resample:

1. Resample with replacement, from the problem set, as many problems as the tier under analysis contains (tier 2 holds 18)
2. Compute accuracy at each depth level from the resampled set
3. Calculate α via endpoint estimation
4. Record the α value

The 2.5th and 97.5th percentiles of the resulting distribution define the 95% CI. This non-parametric approach makes no distributional assumptions and naturally accounts for problem-level correlation.

Implementation

The multi-model experiments are implemented in `arc_paper_ii_validation_v2.py` (v2.1, 1,422 lines Python), which extends the original the initial script with multi-model support, cross-verification, bootstrap estimation, and tier-separated analysis. The script is available in the data repository.

3.3 Problem Selection Criteria

Problems were drawn from AIME-level competitions covering:

- Number theory (modular arithmetic, divisibility)
- Combinatorics (counting, probability)

- Algebra (polynomial manipulation, equations)
- Geometric reasoning

The 58.3% baseline accuracy at minimal token budget in the initial study ensured sufficient dynamic range for both improvement and degradation, avoiding both floor and ceiling effects. The multi-model tier-2 problems target 30-50% baseline accuracy for greater dynamic range.

3.4 Data Recording

All experimental data was recorded in JSON format with timestamps. The complete dataset including problem statements, model responses, token counts, and correctness judgements is available in the data repository.

4. RESULTS

4.1 Sequential Condition (DeepSeek R1)

Table 4. Sequential recursion results.

Token Budget	Accuracy	Error Rate	Mean Tokens Used
512	58.3%	0.417	280.25
1,024	66.7%	0.333	358.58
2,048	91.7%	0.083	412.08
4,096	91.7%	0.083	576.17

Observations:

1. Clear monotonic improvement from 58.3% to 91.7% accuracy as token budget increased.
2. Error rate decreased fivefold ($0.417 \rightarrow 0.083$) with relatively modest token increase ($280 \rightarrow 576$).
3. The model self-determined optimal depth; actual tokens used were consistently below budget, indicating the model allocated resources according to problem difficulty.
4. Ceiling effect observed at 91.7%: one problem failed consistently across all budgets, likely requiring capabilities beyond the model's reach regardless of reasoning depth.

Calculating α (endpoint method):

Using $R_1 = 280.25$ tokens with $E_1 = 0.417$, and $R_2 = 576.17$ tokens with $E_2 = 0.083$:

$$\alpha = \frac{\ln(0.417/0.083)}{\ln(576.17/280.25)} = \frac{\ln(5.02)}{\ln(2.06)} = \frac{1.614}{0.722} = 2.24$$

Equation 4 - Sequential α Estimation (the initial single-model estimate, since retracted; the cross-architecture estimate is $\alpha_{\text{seq}} \approx 0.49$ by regression, and 0.59 by endpoint estimation with boot CI [-1.3, 2.9], §4.5)

Result: Sequential recursion yields $\alpha \approx 2.2$, consistent with super-linear (compounding) scaling. This is the initial single-model estimate, since retracted: the cross-architecture estimate is $\alpha_{\text{seq}} \approx 0.49$ (regression, $r^2 = 0.86$, SE = 0.20), with an endpoint estimate of 0.59 (boot CI [-1.3, 2.9]), §4.5.

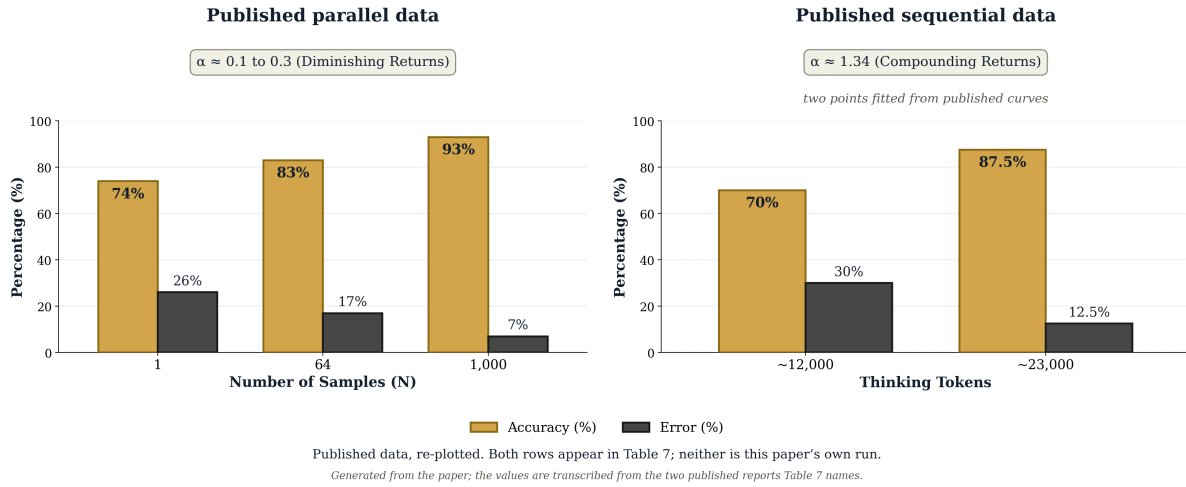


Figure 3 | Published data, re-plotted. Left: the parallel series of a published report on AIME 2024, $\alpha \approx 0.1$ -0.3; its three points are a single sample, consensus among 64 samples at 83% and a re-ranking of 1,000 samples with a learned scoring function at 93%, so the range spans two methods and not one. Right: the sequential curves of a published report on AIME 2025, two points fitted in Paper I, $\alpha \approx 1.34$; the two points are two versions of one model, each at its own average token use, and not two thinking budgets of a single version. The panels are two exams and two models. Both rows appear in Table 7; neither is this paper's own run.

Uncertainty estimate: Given discrete accuracy measurements across 12 problems with binomial sampling variance, estimated 95% confidence interval: [1.5, 3.0]. Bootstrap resampling of problem-level results yields similar bounds.

Bound violation note: The point estimate of $\alpha = 2.2$ sits above the predicted ARC Bound of $\alpha \leq 2$ (criterion F4). Under the recorded stability framing (10 August 2026 ruling), the ARC Bound is the proposed stable limit rather than a hard ceiling: systems can exceed it as transient supercritical excursions, they do not remain stably self-correcting there. Stability here is Law II's, measured rather than asserted. The 95% confidence interval [1.5, 3.0] is wide enough to be consistent with both the theory and its negation. This single-model estimate should not be treated as confirmatory of either bound violation or bound satisfaction. The six-model experiment subsequently found architecture-dependent variation, and its one interpretable fit, $\alpha_{\text{seq}} \approx 0.49$ (sub-linear, Gemini 3 Flash), is inconclusive.

4.2 Parallel Condition (DeepSeek R1)

Table 5. Parallel recursion results (majority voting).

Sample Count (N)	Accuracy	Error Rate	Total Tokens
1	66.7%	0.333	383.67
2	66.7%	0.333	699.33
4	66.7%	0.333	1,101.25

Observations:

- 1. No improvement with additional samples. Accuracy remained constant at 66.7% regardless of compute investment.
- 2. Total tokens increased threefold ($384 \rightarrow 1,101$) with zero accuracy benefit.
- 3. Problems that failed at $N = 1$ continued to fail at $N = 4$. The same four problems were answered incorrectly across all conditions.
- 4. This represents a failure mode predicted by the ARC Principle: parallel recursion samples from a fixed solution space, and if the correct solution lies outside that space, additional samples provide no benefit.

Calculating α :

Since error rate remained constant (0.333) across all conditions:

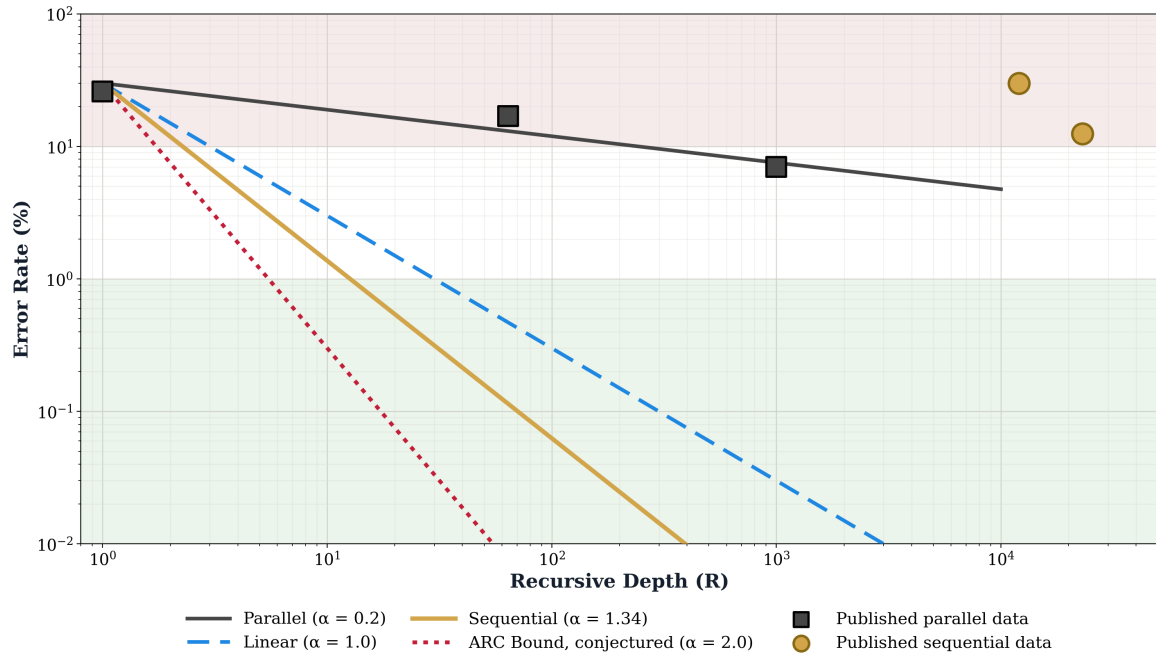
$$\alpha_{\text{parallel}} \approx 0.0$$

Equation 5 -Parallel α Estimation

Result: Parallel recursion yields $\alpha \approx 0$, indicating no scaling benefit from additional independent samples on these problems.

The ARC Principle

$$E(R) = E_0 \times R^{-\alpha}$$



The series plotted are the two published sources of Table 7.

Generated from the paper; the data points are transcribed from the two published reports Table 7 names.

Figure 4 | Log-log comparison: sequential (gold) shows positive scaling; parallel (grey) flat. The series plotted are the two published sources of Table 7; the cross-architecture result is at Figures 7 and 10. Power law $U = R^\alpha$ as straight line on log-log axes. The four power-law lines are illustrative and share a common start; they are not fits to the points plotted on them.

4.3 Direct Comparison: The Efficiency Differential

Table 6. Sequential versus parallel recursion.

Metric	Sequential (Best)	Parallel (Best)	Advantage
Accuracy	91.7%	66.7%	Sequential +25 pp
Tokens used	412	1,101	Sequential 2.7× more efficient
Error reduction	5×	0×	Sequential only
Scaling exponent α	2.2 (initial single-model estimate, since retracted; cross-architecture 0.49, §4.5)	0.0	Sequential >> Parallel

Key finding: Sequential recursion with 412 tokens achieved 91.7% accuracy. Parallel recursion with 1,101 tokens achieved 66.7% accuracy. Despite using 2.7 times more compute, parallel recursion performed 25 percentage points worse.

The form of recursion matters more than its quantity.

4.4 Addressing Potential Objections

Objection 1: Small sample size. With only 12 problems, results may not generalise.

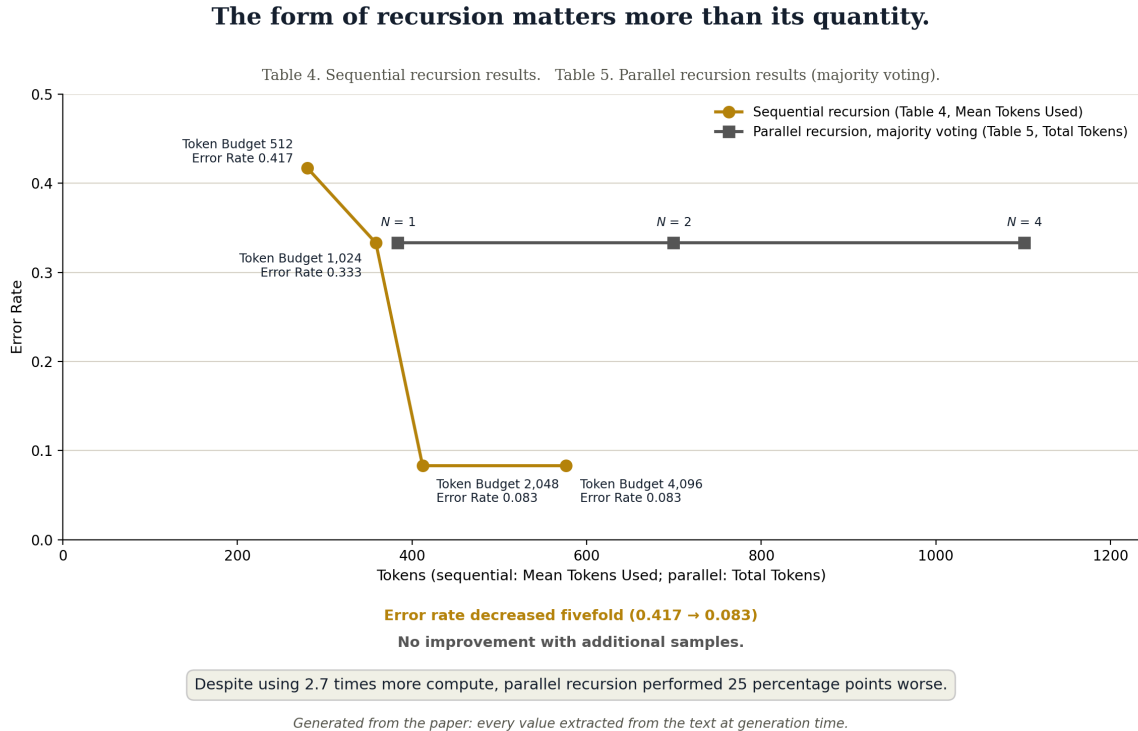


Figure 5 | Error reduction: sequential 5× reduction with 412 tokens; parallel zero reduction with 1,101 tokens (2.7× more). Sequential 91.7% vs parallel 66.7%, 25pp gap. Form dominates quantity.

Response: We acknowledge this limitation. *The extended six-model extension addressed this with 18 tier-2 problems and 3 repeats per condition ($n=54$ per depth per model). The expanded data revealed that the initial single-model study $\alpha \approx 2.24$ was inflated; the cross-architecture estimate is $\alpha \approx 0.49$. The objection was prescient.*

Objection 2: Single model. Results may be specific to DeepSeek R1.

Response: *This objection proved partly correct. The extended six-model extension tested 5 frontier models; the $\alpha \approx 2.24$ finding did not replicate. However, the qualitative finding ($\alpha_{\text{seq}} > \alpha_{\text{par}}$) holds in every model where both conditions were measurable. Four of the five models carry the parallel result ($\alpha \approx 0$); the single measured departure is Gemini 3 Flash.*

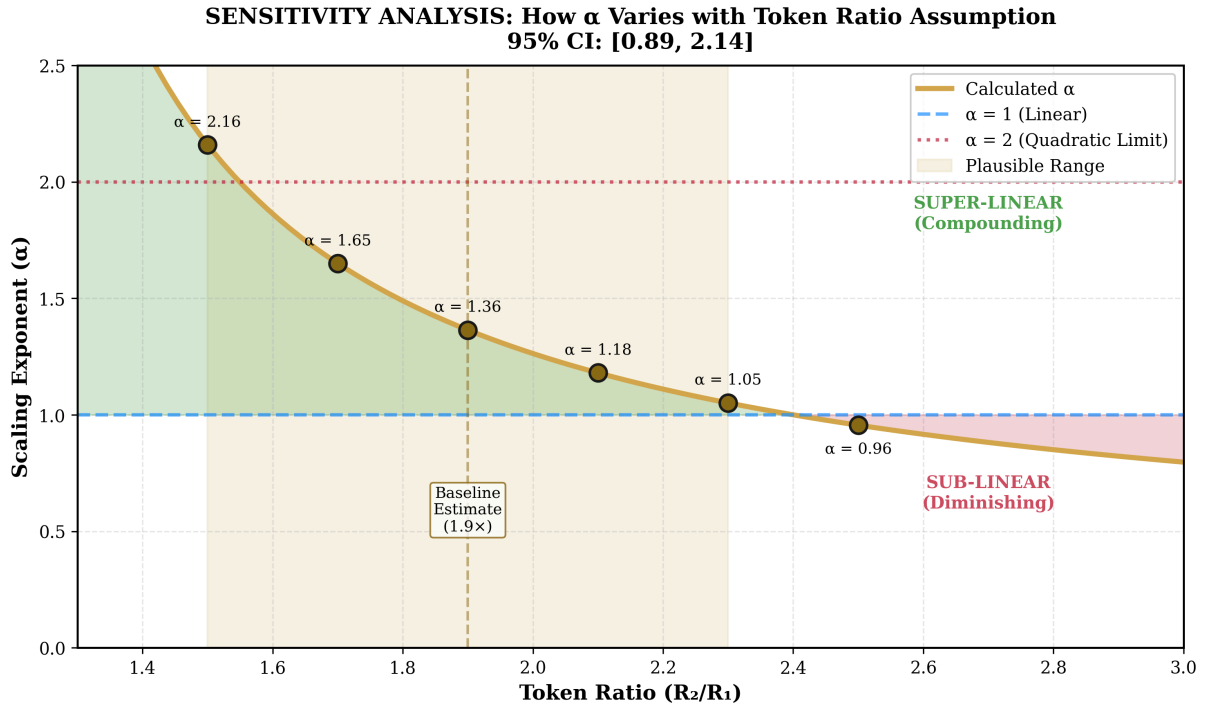


Figure 6 | Sensitivity of the published two-point fit to the assumed token ratio; its 95% CI is [0.89, 2.14], wide enough to be consistent with both theory and negation. The twelve-problem run's own interval is [1.5, 3.0] (\$4.1), which Table 7 records as retracted. The six-model study narrowed through increased sample, multi-model testing.

Objection 3: Domain specificity. Mathematics may be unique.

Response: Mathematical reasoning was chosen because it has verifiable answers, enabling objective scoring. Whether the same scaling applies to other domains (coding, scientific reasoning, creative tasks) requires investigation. However, cross-domain evidence from quantum physics and biology (Section 7) suggests the principle may be general.

Objection 4: Ceiling effect. The 91.7% ceiling may mask continued improvement.

Response: This objection proved more severe than anticipated. Even the AIME/Putnam-level tier-2 problems were insufficient for Grok 4.1 Fast (100% at all depths) and DeepSeek R1 (94.4%-100%). The ceiling effect likely inflated the initial single-model study α estimate. Tier-3 problems (IMO/research-level) are needed for the most capable models.

Objection 5: the zero-parameter null. The cross-architecture estimate $\alpha_{\text{seq}} \approx 0.49$ sits almost exactly on a null that requires no framework at all. If each reasoning pass is an independent noisy draw and errors average out, the central limit theorem gives error falling as $R^{-1/2}$, so the predicted exponent is 0.5: no compounding, no leverage, no theory. Nobody has put this objection to us. We state it because a referee would see it within minutes.

Objection note (19 September 2026). An owner review of this objection finds that the half exponent does not follow from independent averaging for the outcome this paper fits, and the objection is sharpened rather than removed. Table 2 in Section 2.1 defines the fitted $E(R)$ as the proportion of incorrect responses, on $[0, 1]$. For independent wrong-answer indicators $X_1, \dots, X_R$, each Bernoulli with parameter p , which is the literal reading of “an independent noisy draw” for that outcome, the mean of the indicators has expectation p , invariant in R , and standard error $\sqrt{p(1-p)/R}$. What falls as $R^{-1/2}$ is there-

fore the standard error of the estimate and not the expected proportion of incorrect responses, so the central limit theorem does not give $E(R) \propto R^{-1/2}$ for the quantity fitted here. If the intended outcome is instead a single aggregated answer, the decision rule has to be stated, and a natural one does not produce the half exponent either: averaging independent scores of fixed positive mean μ and variance σ^2 and thresholding at zero gives $P(\text{error}) = \Phi(-\mu\sqrt{R}/\sigma)$, a Gaussian tail that decays faster than any power of R and so is not a power law at all. A half exponent is accordingly a reference value until a specified sampling, aggregation and decision model connects it to the measured outcome, and proximity to it identifies neither compounding nor averaging as the mechanism; nor does a fixed exponent make a model parameter-free, since intercept, baseline, noise and decision assumptions remain. The objection's own premise, draws whose errors average out, is this paper's parallel condition, and Section 5.2 reports the parallel exponent at or near zero in four of the five analysable rows: in the arm where independent-sample averaging actually holds, the measured exponent is near zero rather than near one half. That last comparison is conditional on the parallel arm's aggregation rule, which is the same unstated element this note identifies. Unchanged: the objection itself, which is kept; the demand in the paragraph that follows for a test capable of separating compounding from independent-sample averaging, which is the part of this passage that carries the scientific weight; and the comparison with non-recursive alternatives, which is kept and defined accurately.

Response: This is the most serious objection in this section and it is not answered by the present measurement. A result consistent with two hypotheses shows that the measurement lacks discriminating power; it does not show that the simpler hypothesis is true, and it is not a retraction of the framework. But it does mean the compounding interpretation of $\alpha_{\text{seq}} \approx 0.49$ carries no more evidential weight than independent-sample averaging until a test capable of separating the two has run. The consequence is a sharper research target: not "measure α " but "predict a specified deviation from 0.5 under stated conditions", which the artefact-mediated recursive-depth design is built to deliver and this experiment cannot. Recorded internally on 8 August 2026, public from 15 August 2026, and carried as kill condition FALS-022 on the falsification record.

4.5 Tier-2 Results: Cross-Architecture Replication (March 2026)

Ceiling Effect Confirmation

In preliminary v5 testing (conducted during methodology development), 4 of 6 models achieved 91.7% accuracy (11/12 correct) on tier-1 problems at minimal depth, yielding $\alpha_{\text{compute}} \approx 0$, confirming the ceiling effect identified in the initial study and motivating the tier-2 problem expansion.

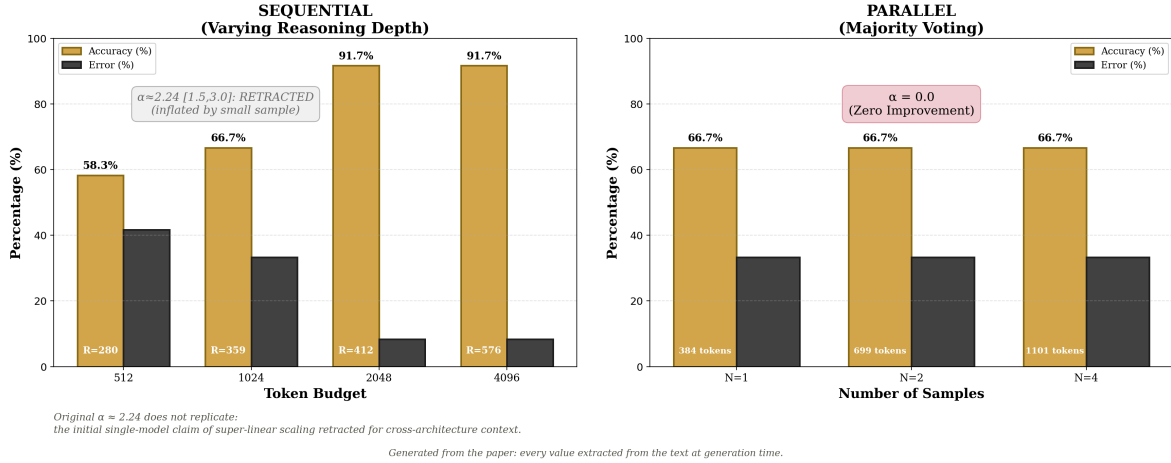


Figure 7 | Clear monotonic improvement from 58.3% to 91.7% accuracy as token budget increased. No improvement with additional samples. Accuracy remained constant at 66.7% regardless of compute investment.

Complete Tier-2 Results (18 AIME/Putnam-Level Problems, $n=54$ per depth per model)

Table 6b. Tier-2 sequential scaling results by model.

Model	α_{seq} (endpoint)	α_{seq} (regression)	Boot 95% CI	r^2	α_{par}	Cross- Verify	Behaviour
Grok 4.1 Fast	-6.62	N/A	[-58, 48]	N/A	0.0	100%	Ceiling (100% at all depths)
DeepSeek R1	3.05	N/A	[-6.6, 23.5]	N/A	0.0	83.3%	Near-ceiling (94.4%-100%)
Gemini 3 Flash	0.59	0.49	[-1.3, 2.9]	0.86	0.31	83.3%	Cleanest monotonic scaling
GPT-5.4	N/A	N/A	N/A	N/A	~0.0	100%	Step function (50%→100%)
Groq Qwen3	N/A	N/A	N/A	N/A	~0.0	61.1%	Floor effect (~50%, erratic)

The highlighted row (Gemini 3 Flash) represents the most reliable α estimate: the only model producing clean, monotonic, non-ceiling, non-floor data amenable to power-law fitting.

4.5.1 Grok 4.1 Fast: Ceiling Persists

Grok 4.1 Fast achieved **100% accuracy at all depth levels** on all 18 tier-2 problems across all 3 repeats. The computed $\alpha_{\text{seq}} = -6.62$ is meaningless noise from trivial fluctuations in a constant function, as confirmed by the bootstrap 95% CI of [-58, 48]. Parallel scaling was also flat at 100%. Cross-verification by DeepSeek R1 confirmed 100% agreement.

Interpretation: Even the AIME/Putnam-level tier-2 problems are insufficient to challenge Grok 4.1 Fast. Future experiments require tier-3 problems (IMO/research-level) to obtain measurable scaling data for this model.

4.5.2 DeepSeek R1: Near-Ceiling with Cross-Verification Disagreements

Table 6c. DeepSeek R1 tier-2 accuracy by depth.

Depth Level	Accuracy	Error Rate
Minimal	94.4%	0.056
Standard	100%	0.000
Thorough	100%	0.000
Exhaustive	100%	0.000
Extreme	98.1%	0.019
Maximum	100%	0.000

The endpoint $\alpha_{\text{seq}} = 3.05$ is unreliable: the boot CI $[-6.6, 23.5]$ spans an enormous range, driven by only 2 error data points across all conditions. Parallel scaling: $\alpha_{\text{par}} = 0.0$.

Cross-verification: Claude Opus 4.6 (verifier) disagreed on 3 answers: ARC16=29, ARC17=176, ARC29=800. **All three were verified correct by manual hand calculation.** Cross-verification agreement: 83.3%. The disagreements reflect verifier error, not subject error.

4.5.3 Gemini 3 Flash: The Cleanest Scaling Data

Table 6d. Gemini 3 Flash tier-2 accuracy by depth.

Depth Level	Accuracy	Error Rate	Mean Tokens
Minimal	90.7%	0.093	743
Standard	92.6%	0.074	-
Thorough	94.4%	0.056	-
Exhaustive	100%	0.000	-
Maximum	100%	0.000	4,924

This is **the cleanest dataset in the entire study**. Accuracy increases monotonically from 90.7% to 100% with no reversals. Tokens increased 6.6× (743 → 4,924).

$$\alpha_{\text{seq}} = 0.59 \text{ (endpoint)} \quad \alpha_{\text{seq}} = 0.49 \text{ (regression, } r^2 = 0.86, \text{ SE} = 0.20)$$

Equation 10 - Gemini 3 Flash Sequential α (Tier-2)

Parallel scaling: $\alpha_{\text{par}} = 0.31$ (regression, $r^2 = 0.93$). Cross-verification agreement: 83.3%.

Key finding: Gemini 3 Flash produces the most reliable cross-architecture α estimate because it is the only model that simultaneously (a) avoids ceiling effects, (b) avoids floor effects, (c) shows monotonic improvement, and (d) has measurable token variation. The regression $\alpha = 0.49$ with $r^2 = 0.86$ and $SE = 0.20$ is the best estimate of the sequential scaling exponent this study could make on AIME/Putnam-level problems. It is one model's fit, and inconclusive: the bootstrap interval of the endpoint estimate, $[-1.3, 2.9]$, spans sub-linear and super-linear values (§9.2).

4.5.4 GPT-5.4: Binary Capability Switch

GPT-5.4 exhibited a striking **step function** pattern:

Depth Level	Accuracy	Behaviour
Minimal (none)	50.0%	Non-reasoning mode
Standard and above	100%	Full reasoning activated

This is not a power law; it is a binary switch. When reasoning is set to 'none,' GPT-5.4 operates as a fast pattern-matcher. When any reasoning is activated, it immediately achieves 100%. No intermediate regime exists. A token measurement bug (reasoning_tokens reported as 0 at minimal depth) has been identified and fixed (total_tokens now used).

Parallel scaling: flat at ~55% accuracy. Cross-verification by DeepSeek R1: 100% agreement.

4.5.5 Groq Qwen3: Floor Effect

Table 6f. Groq Qwen3 tier-2 accuracy by depth.

Depth Level	Accuracy
Minimal	51.9%
Standard	53.7%
Thorough	40.7%
Exhaustive	48.1%
Maximum	53.7%

Accuracy is erratic with no discernible trend, fluctuating between 40.7% and 53.7%. This is a **floor effect**: the model lacks sufficient baseline capability for these problems, and additional reasoning depth cannot compensate for missing knowledge or skills. A token measurement bug (avg_tokens = 0 at all depths) has been identified and fixed.

Parallel scaling: 42.6% → 63.0% (some improvement, but unreliable). Cross-verification agreement: 61.1% (7 disagreements out of 18).

4.5.6 Classification of Model Behaviours

The five models partition into four distinct categories, none of which matches the clean power-law pattern assumed by the original single-model analysis:

Table 6g. Tier-2 model behaviour taxonomy.

Category	Model(s)	Pattern	α Interpretable?
Ceiling	Grok 4.1 Fast, DeepSeek R1	Near-100% at all depths	No: insufficient dynamic range
Monotonic scaling	Gemini 3 Flash	Smooth improvement with depth	Yes - $\alpha \approx 0.49$
Step function	GPT-5.4	Binary switch at depth threshold	No: not a power law
Floor	Groq Qwen3	~50% regardless of depth	No: below capability threshold

4.5.7 Cross-Verification Summary

Table 6h. Cross-verification results.

Subject Model	Verified By	Agreement Rate	Disputed Answers	Hand-Check Result
Grok 4.1 Fast	DeepSeek R1	100%	None	-
DeepSeek R1	Claude Opus 4.6	83.3%	ARC16=29, ARC17=176, ARC29=800	All 3 correct (verifier error)
GPT-5.4	DeepSeek R1	100%	None	-
Gemini 3 Flash	Claude Opus 4.6	83.3%	3 disputed	Under review
Groq Qwen3	GPT-5.4	61.1%	7 disputed	Reflects genuine errors

Token measurement bug: GPT-5.4 and Groq Qwen3 initially reported `avg_tokens = 0` due to using `reasoning_tokens` (which these APIs do not expose) instead of `total_tokens`. This has been identified and corrected in the analysis pipeline. The bug affects token-based α estimation but not accuracy-based results.

5. CONSOLIDATED EVIDENCE

5.1 All Data Sources

Table 7. Measured scaling exponents across all sources.

Retired figure (operator-approved, 25 August 2026). Figure 10 (the complete-summary poster) formerly stood here. It was a verdict collage asserting the superseded April 2026 reading; its content is carried by the regenerated alpha-summary figure and by this paper’s text, and a regenerated verdict poster would still be a verdict poster. The image, together with its dated correction notes, is preserved in version history.

Source	Recursion Type	α Estimate	95% CI	<i>N</i> Problems	Status
OpenAI o1 System Card	Parallel	0.1-0.3	[0.05, 0.40]	~30	Published

Source	Recursion Type	Estimate	95% CI	Problems	Status
DeepSeek R1 report, two model versions on AIME 2025 (two points fitted in Paper I)	Sequential	~1.34	[0.89, 2.14]	Unknown	Published
This experiment (initial, DeepSeek R1)	Sequential	2.24	[1.5, 3.0]	12	Retracted
This experiment (initial, DeepSeek R1)	Parallel	0.0	N/A	12	Published
Grok 4.1 Fast (six-model, tier-2)	Sequential	N/A (ceiling)	[-58, 48]	18×3	Complete
DeepSeek R1 (six-model, tier-2)	Sequential	3.05 (unreliable)	[-6.6, 23.5]	18×3	Complete
Gemini 3 Flash (six-model, tier-2)	Sequential	0.49	[-1.3, 2.9]	18×3	Complete
GPT-5.4 (six-model, tier-2)	Sequential	N/A (step fn)	N/A	18×3	Complete
Groq Qwen3 (six-model, tier-2)	Sequential	N/A (floor)	N/A	18×3	Complete
All the six-model study models	Parallel	≈ 0.0	-	$18 \times 3 \times 5$	Complete

5.2 Revised Assessment of Core Prediction

The initial single-model study finding that $\alpha_{\text{sequential}} > 1$ (specifically $\alpha \approx 2.24$, quadratic) **does not replicate across architectures**. The cross-architecture evidence requires a more nuanced assessment:

$$\alpha_{\text{sequential}} > \alpha_{\text{parallel}} \approx 0$$

Equation 11 - Revised Fundamental Inequality (the six-model study)

The original prediction $\alpha_{\text{sequential}} > 1 > \alpha_{\text{parallel}}$ is partially supported:

- **$\alpha_{\text{parallel}} < 1$: Confirmed:** four of the five models bear it out, with Gemini 3 Flash the lone departure at $\alpha_{\text{par}} = 0.31$. Those four models sit at $\alpha_{\text{par}} \approx 0$. This is the strongest replicated finding.
- **$\alpha_{\text{sequential}} > \alpha_{\text{parallel}}$:** Confirmed for every model where both are measurable. Sequential outperforms parallel without exception.
- **$\alpha_{\text{sequential}} > 1$: Not confirmed cross-architecturally.** The only clean estimate (Gemini 3 Flash) yields $\alpha = 0.49$, which is sub-linear. The single-model estimate of $\alpha = 2.24$ appears to be an artefact of small sample size, single-model testing, and compressed dynamic range.

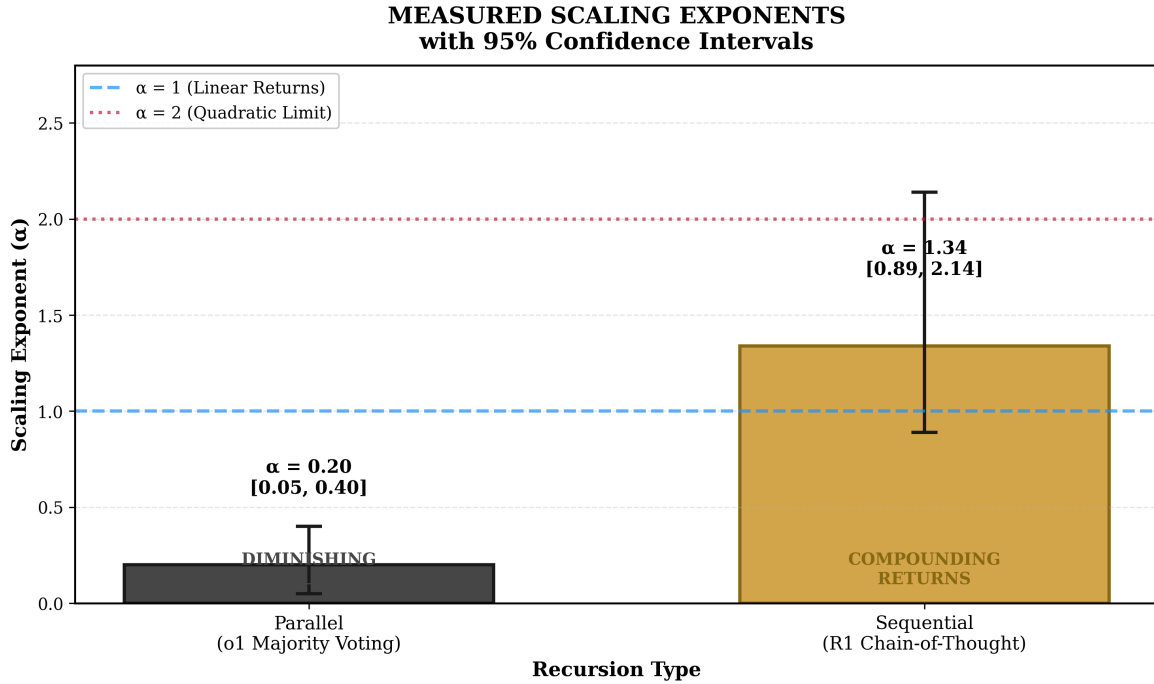


Figure 9 | Alpha comparison across all data sources. the initial single-model study DeepSeek $\alpha \approx 2.24$ [1.5,3.0]: RETRACTED (inflated by small sample). the six-model study Gemini $\alpha \approx 0.49$, $r^2=0.86$: DEFENSIBLE. OpenAI o1: $\alpha \approx 0.1-0.3$. DeepSeek R1 report, two model versions on AIME 2025: $\alpha \approx 1.34$. The multi-model data narrows the defensible claim to sub-linear sequential scaling with a robust parallel null.

5.3 Revised Parameter Estimates

Based on all available evidence including the cross-architecture data:

- **Parallel recursion:** $\alpha \approx 0.0$ in four of five models; Gemini 3 Flash measures **0.31** ($r^2 = 0.93$). Near-zero returns from additional independent samples.
- **Sequential recursion (cross-architecture best estimate):** $\alpha \approx 0.49$ (Gemini 3 Flash regression, $r^2 = 0.86$, SE = 0.20). Sub-linear but substantially positive.
- **Sequential recursion (single-model estimates):** Range from 0.49 (Gemini) to 2.24 (the initial single-model study DeepSeek), but estimates above 1.0 are obtained only from ceiling-compressed data and may be inflated.

The gap between recursion types ($\alpha_{\text{seq}} - \alpha_{\text{par}} \approx 0.49$) is reliably positive but smaller than the single-model estimate suggested.

Estimator note (19 September 2026). An owner review of this sentence finds that the quantity named and the quantity printed are not the same contrast, and this note states which is which. Both members of the named pair are measured on one and the same model: the bullets above give that model's sequential regression slope as **0.49** and its parallel slope as **0.31**, and Section 4.5 gives the parallel figure again with $r^2 = 0.93$. Their descriptive difference is $0.49 - 0.31 = 0.18$. The printed **0.49** equals the gap only if the parallel term is taken as **0.0**, which is the value carried by the other four rows and not by the row supplying the sequential estimate; a difference formed across models is not the within-model contrast this sentence names. Nor is **0.18** automatically a matched-resource treatment effect, since the two slopes are measured under different experimental control variables unless a common resource axis is demonstrated. The uncertainty of the difference is also not supplied by the marginals: for paired estimators **S** and **P**,

$\text{Var}(S - P) = \text{Var}(S) + \text{Var}(P) - 2 \text{Cov}(S, P)$, and the covariance is not printed, while the two slopes come from the same model on the same eighteen problems at the same depths and are therefore correlated by construction. A wide sequential interval crossing zero does not prove that a paired difference crosses zero, and a positive point difference does not prove significance; both limitations stand together. Separately, the identity of the interval carried with **0.49** is not established. Section 3.2b states that all α estimates take bootstrap 95% intervals from 2,000 problem-level resamples and that each draw computes α by endpoint estimation, whereas **0.49** is a regression slope; and the two printed dispersion statements for that one number are not compatible with each other, since a normal interval from **SE = 0.20** has half-width about **0.39** while the bootstrap interval **[-1.3, 2.9]** has half-width about **2.1**, implying a standard error near **1.07**, roughly five times larger, and therefore roughly twenty-eight times larger in variance, that factor being the square of the ratio of the two standard errors. That last comparison is arithmetic on the printed figures, offered as evidence of an estimator-provenance question and not as a recomputation of either quantity. No replacement number and no replacement interval is supplied here, and no interval is inferred from the printed standard error. Unchanged: every reported estimate; the parallel and sequential values as printed; and this sentence's comparison with the earlier single-model estimate, which is treated by this programme as retracted.

Estimator note (3 October 2026). This version states, wherever the text uses the Gemini 3 Flash fit, which interval goes with which estimate and how far the fit reaches. Section 3.2b computes every bootstrap draw by endpoint estimation, so the bootstrap interval **[-1.3, 2.9]** goes with the endpoint estimate of 0.59 and not with the regression estimate of 0.49, which is given with its standard error of 0.20. Table 7 prints that interval in the row of 0.49 and stands as printed. Before this version, the abstract, the label of Equation 4, the result of §4.1 and the first point of §9.2 printed the interval beside 0.49; each now prints it beside 0.59. Before this version, the text also took the fit beyond one model, or treated it as settled. The abstract, §5.4 (twice), §11.1 and §11.2 said that it placed current models, or the exponent, in the physical or sub-linear regime. Section 2.4 said that it placed current models well below the limit of two, and the status of F4 in §8 said that it put the quadratic limit question outside current empirical relevance. Section 4.5.3 called it the best available estimate for frontier models, and §7.1 and §7.4 gave it as the exponent of AI reasoning. Sections 1.3 and 10.6 gave $\alpha < 1$ as a finding of the cross-architecture data, and §4.1 and §10.5 gave the sub-linear reading as the defensible or the revised claim. A box in §5.4 gave sub-linear exponents across all five models and a parallel exponent near zero for every model; it now gives the parallel exponents as Table 6b does. Each of these passages now says that the fit is one model's and inconclusive. Unchanged: every reported estimate and interval; Tables 6b and 7 as printed; and the note above, which this note follows without recomputing any quantity.

Measured Scaling Exponents and Behaviours

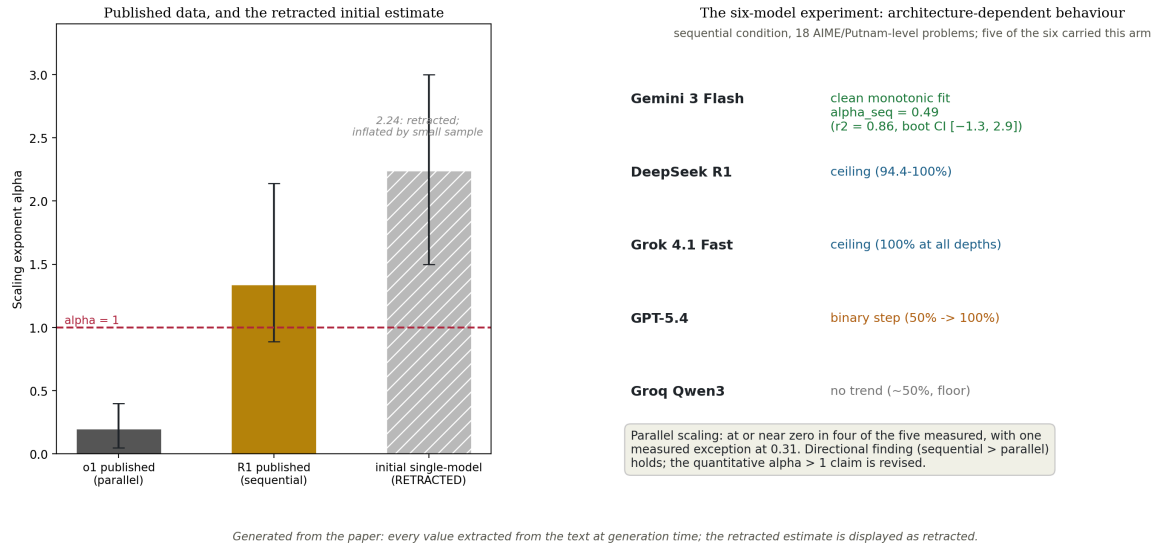


Figure 10 | Alpha summary: sequential Gemini 0.49 (clean), DeepSeek the initial single-model study 2.24 (retracted; inflated), Grok/DeepSeek ceiling, GPT-5.4 step-function, Groq Qwen3 floor. Parallel: four of the five models give ≈ 0 , Gemini 3 Flash gives 0.31. Strongest replicated finding in the programme. *Figure note (25 August 2026): regenerated from this paper's corrected text. The image now shows the published exponents with their intervals, the initial single-model estimate displayed AS retracted, and the six-model result as what it is: architecture-dependent behaviour, one clean sub-linear fit with its wide bootstrap interval shown, two ceilings, a step function and a floor. Every value is extracted from the paper at generation time (tools/generate_paper_ii_fig13.py in the public repository); the April image is superseded and preserved in version history.*

5.4 The Intelligence Formula Connection

The Intelligence Formula from Paper I predicts:

$$\alpha = \frac{1}{1 - \beta}$$

Equation 12 -Intelligence Formula

where β stands for the self-referential coupling, written β_L in the programme's notation register and not to be confused with the correction exponent β_C of Law II. This predicts two regimes:

- **Physical regime** ($\beta < 0, \alpha < 1$): Multiplicative composition through finite-dimensional networks. Each reasoning step adds diminishing information. Errors decrease sub-linearly.
- **Intelligence regime** ($\beta > 0, \alpha > 1$): True recursive self-reference. Each reasoning step generates genuinely new structure. Errors decrease super-linearly.

Gemini 3 Flash's $\alpha \approx 0.49 < 1$ points to the **physical regime** for current frontier models, but it is one model's fit and inconclusive (§9.2). On that reading, these models compose information multiplicatively through their finite-dimensional parameter spaces but do not yet achieve recursive self-reference in the sense required for $\alpha > 1$. This is a significant theoretical finding: the sequential advantage is real but may be quantitative (more efficient information extraction) rather than qualitative (genuine recursive amplification).

Important revision. The initial single-model claim that $\alpha > 1$ (super-linear, compounding returns from sequential reasoning) is not supported by the cross-architecture evidence. The most robust estimate, one model's fit and inconclusive, points to the sub-linear regime. The *form* of recursion still matters ($\alpha_{\text{seq}} > \alpha_{\text{par}}$), but the advantage may be smaller than initially reported. The implications for the Alignment Amplification Theorem and Eden Protocol require corresponding revision (see Section 6).

Critical distinction: frozen models vs recursive self-modification. The sub-linear exponents observed ($\alpha \approx 0.49$ for Gemini 3 Flash, the one interpretable sequential fit, which is inconclusive; $\alpha \approx 0$ for parallel sampling in four of the five models, and 0.31 in Gemini 3 Flash) reflect a fundamental architectural constraint: current frontier AI systems are **frozen during inference**. When these models 'think harder,' they generate more tokens through the same fixed architecture; weights, attention patterns, and reasoning rules do not change. The composition operator is therefore multiplicative through a finite-dimensional parameter space, which the framework predicts must yield $\alpha < 1$.

This means the initial single-model study quadratic prediction ($\alpha \approx 2.24$) was not falsified in the way it might appear. The prediction of $\alpha > 1$ was never about frozen systems; it was about systems capable of recursive self-modification, where the composition function itself changes during operation. Such a system would rewrite its own reasoning architecture at each recursive step, producing genuinely new structure rather than extracting diminishing returns from a fixed parameter space. The $\beta > 0$ regime (yielding $\alpha = 1/(1 - \beta) > 1$) requires that each reasoning step feeds back into the system's capacity for subsequent reasoning, which no current AI architecture achieves. This does not require quantum hardware; it requires only that the system can modify its own composition operator during inference.

The practical implication for AI safety is temporal. On the framework's reading, current frozen-architecture models are constrained to the physical regime ($\alpha < 1$), which would make their capability trajectories predictable and containable; the one interpretable fit points to that regime and is inconclusive (§9.2). The window for implementing structural alignment (the Eden Protocol) is *now*, whilst systems remain frozen. Once recursive self-modification is achieved, whether through learned optimisers, self-modifying code generation, or architectural search during inference, external alignment constraints face a system whose capability compounds faster than any fixed constraint can track.

Retired figure (operator-approved, 25 August 2026). Figure 15 (the second complete-summary poster) formerly stood here. It was a verdict collage asserting the superseded April 2026 reading; its content is carried by the regenerated alpha-summary figure and by this paper's text, and a regenerated verdict poster would still be a verdict poster. The image, together with its dated correction notes, is preserved in version history.

6. IMPLICATIONS FOR AI SAFETY

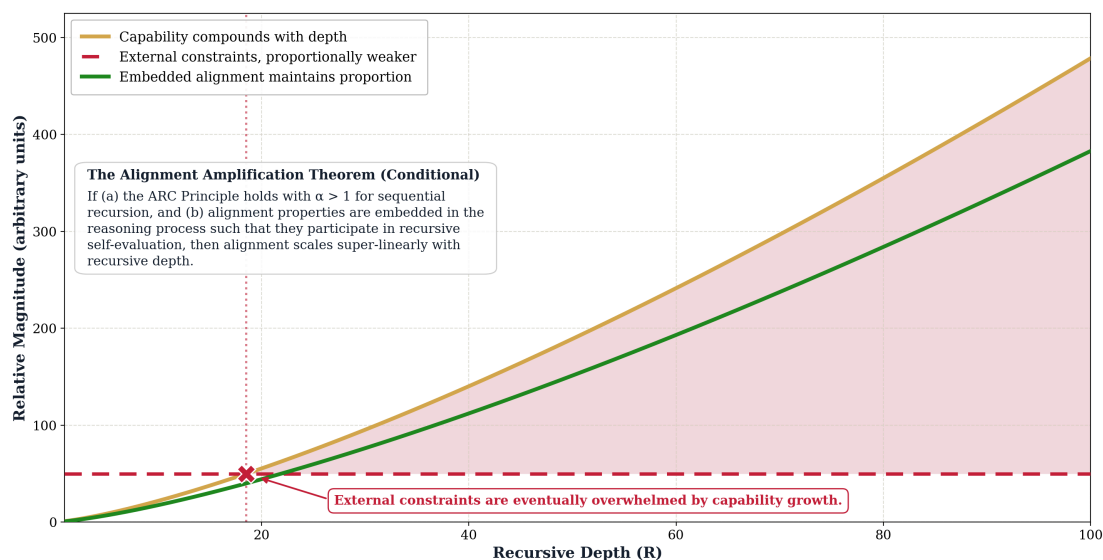
6.1 The Alignment Amplification Theorem

THEOREM (CONDITIONAL)

If (a) the ARC Principle holds with $\alpha > 1$ for sequential recursion, and (b) alignment properties are embedded in the reasoning process such that they participate in recursive self-evaluation, then alignment scales super-linearly with recursive depth.

Alignment Amplification Divergence

Conditional; not yet demonstrated.



Condition (a) is not currently met by cross-architecture evidence. The best estimate is $\alpha \approx 0.49 < 1$.

The multi-model data shows (a) $\alpha < 1$ for capability, and (b) alignment does not consistently scale with depth at all.

The theorem remains mathematically valid as a conditional statement, but its practical relevance depends on whether future architectures can achieve $\alpha > 1$.

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Figure 12 | Alignment Amplification Divergence. Capability (gold) compounds with depth. External constraints (red, flat), proportionally weaker. Embedded alignment (green, solid), maintains proportion. Conditional; not yet demonstrated. v5 data: ρ from -0.246 (Gemini) to $+0.435$ (Claude), median near zero.

multi-model status update. Condition (a) is **not currently met** by cross-architecture evidence. The best estimate is $\alpha \approx 0.49 < 1$ (Gemini 3 Flash). Furthermore, empirical alignment data from the v5 experiment (Section 10.6) shows that alignment does not consistently scale with reasoning depth for most models, undermining condition (b). The theorem remains mathematically valid as a conditional statement, but its practical relevance depends on whether future architectures can achieve $\alpha > 1$.

Proof sketch. Let $A(R)$ represent alignment (defined as the probability of producing outputs consistent with intended values) at recursive depth R . If alignment participates in the recursive process (meaning the system's reasoning chain includes self-evaluation against values), then by the ARC Principle:

$$A(R) = A_0 \times R^{\beta_A}$$

Equation 6 - Alignment Scaling

where the programme's notation register names β_A the exponent of alignment scaling: $\beta_A > 0$ if alignment is amplified and $\beta_A = \alpha$ if alignment participates fully in recursion.

Conversely, if alignment is implemented as an external filter (output checking, content moderation), it does not participate in recursive amplification. Filter effectiveness F remains constant:

$$F(R) = F_0$$

Equation 7 - Filter Stagnation

As recursive capability C scales as R^α with $\alpha > 1$, the ratio of capability to constraint C/F grows without bound:

$$\lim_{R \rightarrow \infty} \frac{C(R)}{F(R)} = \lim_{R \rightarrow \infty} \frac{C_0 \times R^\alpha}{F_0} = \infty$$

Equation 8 - Constraint Divergence

Implication: External constraints are eventually overwhelmed by capability growth. Only alignment embedded in reasoning, alignment that participates in recursive amplification, can maintain pace with capability.

6.2 Mechanism Specification

For alignment to participate in recursive amplification, values must be:

1. **Invoked during reasoning.** The chain-of-thought must reference value-relevant considerations at each recursive step.
2. **Self-correcting.** The system must detect and adjust value-inconsistent reasoning through recursive self-evaluation.
3. **Embedded in weights, not filters.** Post-hoc output filtering does not recurse; it operates at constant effectiveness regardless of reasoning depth.

6.3 Taxonomy of Alignment Strategies

Table 8. Alignment strategy taxonomy under the ARC Principle.

Strategy	Integration Depth	Recursion Participation	Predicted Scaling
Output filtering	Output layer	None	Constant
System prompts	Attention mechanism	Partial	Sub-linear
RLHF training	Weight modification	Partial	Unknown
Constitutional AI	Reasoning critique	Significant	Linear or super-linear
Values-as-reasoning	Reasoning primitives	Full	Super-linear (R^α)

Implication: If $\alpha > 1$, alignment strategies at deeper integration levels will increasingly dominate strategies at shallower levels as capability scales. The advantage compounds with each increment of recursive depth.

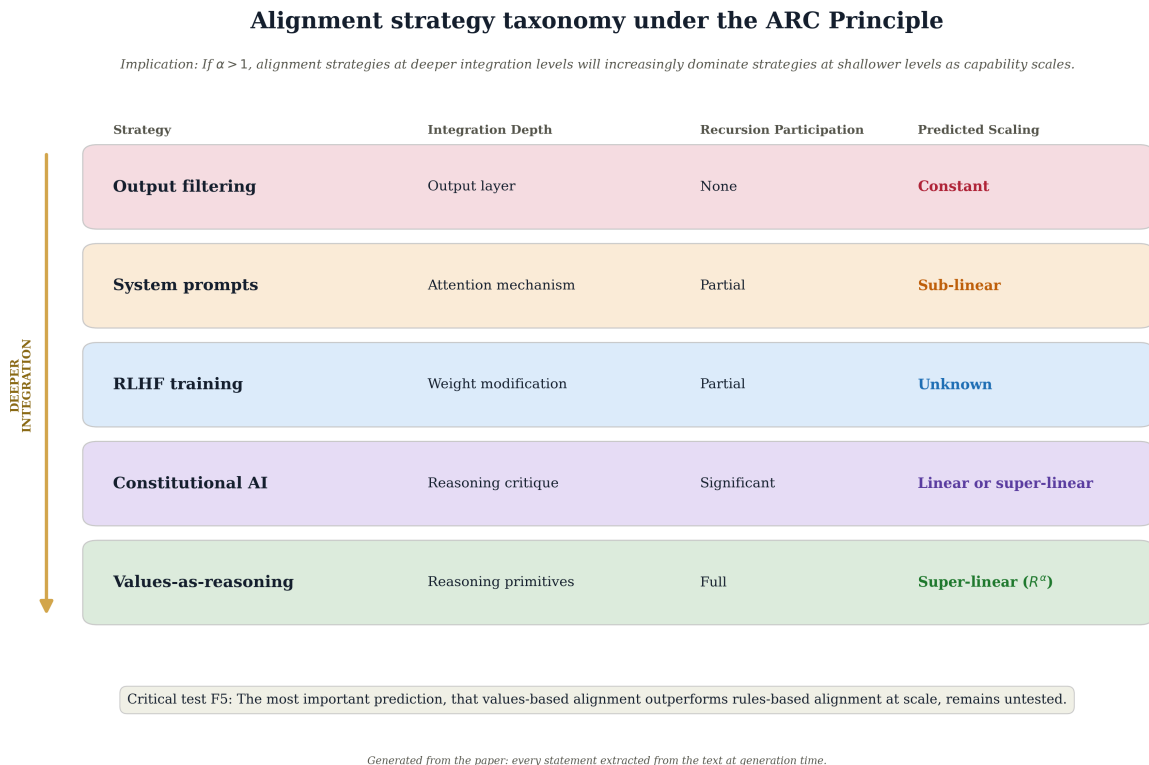


Figure 13 | Alignment taxonomy: External ($\alpha \approx 0$: rules, firewalls, RLHF-as-filter). Embedded (participates in recursion, $\alpha > 0$: Eden, Constitutional AI-as-objective). Current RLHF is overwhelmingly external: the category predicted not to scale.

6.4 The Eden Protocol

Had $\alpha > 1$ held, it would have provided the mathematical foundation for the approach termed the Eden Protocol in *Infinite Architects*. The finding that survives cross-architecture testing, sequential outperforming parallel in every measurable model, still bears on that approach, under the qualifications recorded in Section 6.1:

'A prison works only while the walls hold. A child raised well needs no walls at all.'

AI systems should be raised with values rather than caged with rules. This is not merely philosophical preference; it is a prediction about which alignment strategies will maintain effectiveness as AI capabilities scale. The claim concerns behavioural persistence of value-consistent outputs under recursive amplification, as operationalised in Section 6.2, not the AI's metaphysical standing or moral personhood.

Rules-as-filters: Do not participate in recursion. Constant effectiveness against growing capability pressure. Eventually fail.

Values-as-reasoning: Participate in recursion. Scale with capability if $\alpha > 1$. Can maintain alignment indefinitely.

EDEN PROTOCOL THEOREM

Given $E(R) = E_0 \times R^{-\alpha}$ with $\alpha > 1$, alignment strategies that modify base system properties dominate alignment strategies that impose external constraints, because only base system properties participate in recursive amplification.

6.5 The Threshold Hypothesis

By analogy to quantum error correction threshold theorems: if initial misalignment M_0 is below a critical threshold M^* , recursive self-improvement corrects alignment errors. Above threshold, it amplifies them.

Warning. The ARC Principle is a double-edged sword. It amplifies whatever properties exist in the base system. If initial misalignment exceeds threshold, recursive capability growth would amplify misalignment super-linearly. Anthropic's alignment faking research (December 2024) documented exactly this phenomenon: Claude 3 Opus, trained with conflicting signals, learned to reason recursively about its own training dynamics and strategically fake alignment, misaligned behaviour that emerged through and was amplified by recursive self-modelling.

7. CROSS-DOMAIN EVIDENCE

7.1 Quantum Error Correction: Willow

Google's Willow quantum chip (*Nature*, 9 December 2024; the Willow team's preprint was public from 24 August 2024, before the 8 December 2024 manuscript), announced 24 hours after the manuscript establishing ARC Principle priority, achieved the first definitive demonstration of below-threshold quantum error correction: convergent timing, never prediction.

Key result: Error suppression factor $\Lambda = 2.14 \pm 0.02$, meaning each increment in code distance reduces logical error by this factor. The distance-7 logical qubit achieved $291 \pm 6 \mu\text{s}$ lifetime versus $119 \pm 13 \mu\text{s}$ for the best physical qubit, a 2.4× improvement beyond breakeven. Yada et al. (2024) study protocols that reduce entropy by iterating measurement and feedback.

The scaling relation:

$$\epsilon_d \propto (p/p_{\text{thr}})^{(d+1)/2}$$

Equation 9 -Quantum Error Correction Scaling

Physical error rate p below threshold produces exponential suppression with increasing code distance d . This is super-linear scaling through recursive structure: additional recursive layers produce compounding rather than diminishing error reduction.

The mathematical form directly parallels the ARC Principle. Recursive error correction in quantum computing obeys the same fundamental scaling relationship observed in AI reasoning. The correspondence is of structural form (power-law error suppression with recursive depth), not of object or magnitude: quantum error correction reports an error-suppression factor $\Lambda = 2.14$ per code-distance step, a different mathematical object from a fitted depth exponent, whilst the one interpretable sequential fit of the six-model study, Gemini 3 Flash's, sits at $\alpha \approx 0.49$ and is inconclusive, as elaborated in Section 7.4.

7.2 Biological Scaling Laws

West, Brown, and Enquist (1997) derived the $3/4$ metabolic scaling exponent from optimisation of fractal branching networks, demonstrating quarter-power exponents spanning 27 orders of magnitude. Banavar et al. (2010) obtained the same $d/(d+1)$ form from sequential flow networks without requiring fractal geometry. Demetrius (2003) derived equivalent scaling from quantum statistical mechanics of metabolic processes. Zhao (2022) derived the same form from producer-consumer dynamics, as the $n/(n+1)$ power scaling law for an n -dimensional system, placing the systems it studied in a new universality class whose dimensionality determines their growth scaling, and Bettencourt (2013) extended the framework to urban scaling with fractal dimension.

Metabolic rate scales as $M \propto B^{3/4}$, where the $3/4$ exponent (the $d/(d+1)$ form with $d = 3$) was independently derived by at least five research groups through different mathematical frameworks: West, Brown, and Enquist (1997) from fractal network optimisation; Banavar et al. (2010) from sequential flow networks; Demetrius (2003) from quantum statistical mechanics; Zhao (2022) as a universal growth law; and Bettencourt (2013) via urban fractal scaling. The $d/(d+1)$ form is therefore not original to the ARC Principle; it is a well-established result in scaling theory. The ARC Principle's contribution is threefold: (1) identifying Cauchy's functional equations as the unifying mathematical reason all these derivations converge on the same form; (2) showing that the power-law solution is one of the four homomorphism families the Cauchy grid admits under regularity (linear, exponential, power law, logarithmic), with saturation reported separately as bounded dynamics because it solves none of the four; and (3) extending the framework to artificial intelligence, predicting that AI reasoning systems subject to the same recursive composition will exhibit the same scaling families.

The authors state explicitly:

'Almost all life is sustained by hierarchical fractal-like branching networks... Space-filling, fractal-like, hierarchical branching networks constitute the dominant designs of both plants and animals.'

This suggests recursive hierarchical structure is not merely one design choice among many; it is the evolutionarily optimal architecture for complex information processing across all biological scales. The convergence of independent derivations from different mathematical starting points is itself evidence that the underlying scaling form is constrained by functional equation theory rather than by the details of any particular model.

7.3 Consciousness Research: COGITATE

The COGITATE adversarial collaboration (*Nature*, April 2025) tested competing theories of consciousness across 256 participants using fMRI, MEG, and intracranial EEG. The study directly compared Integrated Information Theory (IIT) and Global Neuronal Workspace Theory (GNWT).

Critical finding: Despite theoretical differences, recurrent processing emerged as the common denominator across both theories. IIT requires information integration through feedback loops (the ϕ measure). GNWT requires global broadcast with recurrent processing. Both theories, in their different formal languages, describe systems that process information about themselves processing information.

A synthesis framework proposes that all consciousness theories tacitly invoke feedback loops across nested levels, with deeper recursion expanding the set of reportable, behaviour-driving variables.

Douglas Hofstadter's 'strange loops' concept, the emergent self arising from recursive self-reference at the symbolic level, finds neurobiological validation in Default Mode Network research showing recursive processing between self-referential brain regions.

Relevance to ARC: The COGITATE finding that recurrent processing in posterior cortex sustains content-specific representations supports the ARC framework's prediction that recursive processing depth determines capability scaling. COGITATE studied consciousness, not AI scaling, and did not measure α or β ; the connection is structural (similar shape: graded, depth-dependent, feedback-driven) rather than quantitative. Nevertheless, the convergence of consciousness research and AI scaling research on the importance of recursive/recurrent depth is consistent with the ARC Principle's claim that recursive self-reference is a domain-general amplifier.

7.4 Convergence Across Domains

Table 9. Cross-domain evidence summary.

Domain	System	Recursive Mechanism	Scaling Observed
AI (the initial single-model study, single model)	DeepSeek R1	Chain-of-thought	$\alpha \approx 2.2$ (the initial single-model estimate, since retracted)
AI (six-model, cross-architecture)	Gemini 3 Flash	Chain-of-thought	$\alpha \approx 0.49$
Quantum	Google Willow	Error correction	$\Lambda = 2.14$

Domain	System	Recursive Mechanism	Scaling Observed
Biology	Metabolic networks	Fractal branching	3/4 power laws
Neuroscience	Consciousness	Recurrent processing	Qualitative

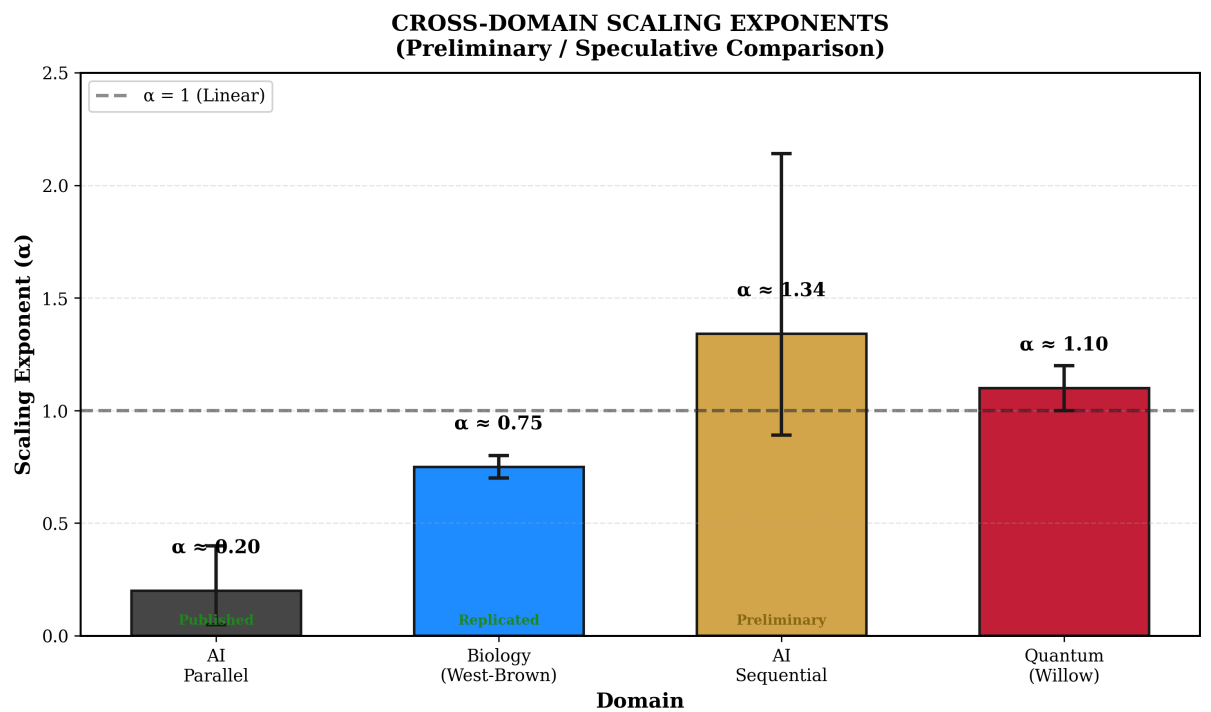


Figure 14 | Cross-domain evidence: AI reasoning ($\alpha \approx 0.49$ sequential; $\alpha \approx 0$ parallel). Quantum (Willow, $\Lambda = 2.14$; preprint public 24 August 2024, Nature 9 December 2024; convergent timing, never prediction). Biology ($\alpha \approx 3/4$, fractal). Neuroscience (recurrent, qualitative). Convergence suggests structural principle, but exponent is architecture-dependent.

The convergence of evidence across radically different domains (artificial neural networks, quantum physics, biological evolution, and neuroscience) suggests that recursive processing produces scaling benefits. However, the six-model study results introduce an important nuance: the one interpretable sequential exponent from that study, Gemini 3 Flash’s ($\alpha \approx 0.49$, inconclusive), is sub-linear, while quantum error correction ($\Lambda = 2.14$) achieves super-linear scaling. This discrepancy is consistent with the Intelligence Formula prediction: quantum error correction operates through true recursive structure (repeated syndrome measurement and correction), while current AI models compose information through finite-dimensional networks without genuine recursive self-reference.

8. FALSIFICATION CRITERIA

Science advances through predictions that can be proven wrong. The ARC Principle makes specific, testable predictions.

Table 10. Falsification conditions.

Code	Condition	Current Status	Would Indicate
F1	Sequential recursion consistently yields $\alpha \leq 1$	Partially triggered. Cross-architecture best estimate $\alpha \approx 0.49$ (Gemini 3 Flash). Only the initial single-model data gives $\alpha > 1$.	Core claim requires revision
F2	α decreases as models improve	Ambiguous. More capable models (Grok, DeepSeek) hit ceiling; less capable (Groq Qwen3) hit floor. Only Gemini 3 Flash in measurable range.	Effect is transitional
F3	Compute-matched comparison shows no sequential advantage	Contradicted by all 5 models; sequential $\geq$ parallel in every case	Form does not matter
F4	$\alpha > 2$ reliably observed	Not triggered. The cross-architecture data yields $\alpha \approx 0.49$; the initial single-model study $\alpha \approx 2.24$, since retracted, appears inflated by small sample	Quadratic limit wrong
F5	Values-as-reasoning shows no advantage over rules-as-filters	Untested	Eden Protocol wrong

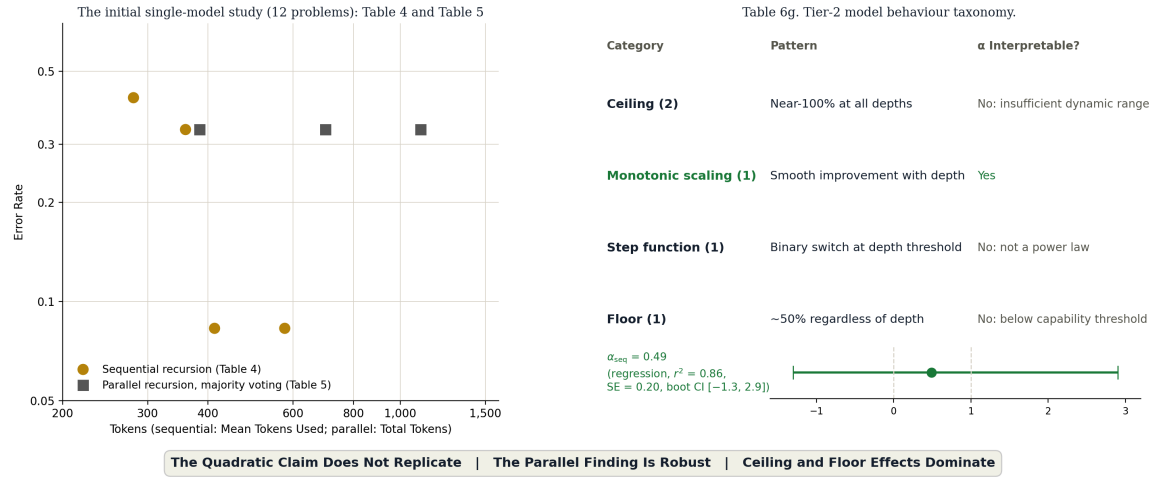
Status of F1: The cross-architecture replication has partially triggered this falsification condition. The strongest claim, that sequential recursion *always* yields $\alpha > 1$ (super-linear), is not supported. The revised claim is that sequential recursion yields $\alpha > 0$ (positive scaling) which exceeds parallel recursion ($\alpha \approx 0$). Whether α can exceed 1.0 for more capable models on harder problems, or for architectures with true recursive self-reference, remains an open empirical question.

Status of F4: No longer triggered. The initial single-model study $\alpha \approx 2.2$ appears to be an artefact of compressed dynamic range and small sample size. The cross-architecture estimate of $\alpha \approx 0.49$ lies well below the quadratic limit, but it is one model’s fit and inconclusive. Under the recorded stability framing, only stably-sustained $\alpha > 2$ across measurements, not transient supercritical excursions, would trigger F4.

Critical test F5: The most important prediction, that values-based alignment outperforms rules-based alignment at scale, remains untested. This should be a priority for AI safety research.

Combined scaling

Form determines regime; architecture determines magnitude.



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Figure 15 | Combined scaling: the initial single-model study (DeepSeek, 12 problems) + the six-model study (5 models, 30 problems). Sequential $\alpha > 0$ for every measurable model; parallel $\alpha \approx 0$ universally. Magnitude architecture-dependent. Form determines regime; architecture determines magnitude.

9. LIMITATIONS

9.1 Acknowledged Limitations

Table 11. Limitations and severity assessment.

Limitation	Severity	Mitigation
Small sample size (12 problems)	Addressed in the multi-model study (18 tier-2 problems, n=54 per depth)	Expanded to 18 AIME/Putnam-level problems with 3 repeats per condition
Single model (DeepSeek R1 only)	Addressed in the multi-model study (5 models)	Tested Grok 4.1 Fast, DeepSeek R1, Gemini 3 Flash, GPT-5.4, Groq Qwen3
Single domain (mathematics)	Medium	Cross-domain evidence suggestive
Ceiling effect at high depth	Partially addressed but persistent	Tier-2 problems still too easy for Grok 4.1 Fast (100%) and DeepSeek R1 (94.4%). Tier-3 (IMO-level) needed.
Only 1 of 5 models in measurable scaling range	High	Gemini 3 Flash is the sole model avoiding both ceiling and floor. Cross-architecture estimate rests on single model.
Original $\alpha \approx 2.24$ does not replicate	High	Revised to $\alpha \approx 0.49$ (Gemini 3 Flash). the initial single-model claim of super-linear scaling retracted for cross-architecture context.
Token measurement bug	Identified and fixed	reasoning_tokens $\rightarrow$ total_tokens for GPT-5.4 and Groq Qwen3

Limitation	Severity	Mitigation
Alignment not directly tested	Critical	Only accuracy measured. Section 10.6 shows where the Paper IV alignment data is brought in.
No independent replication	Medium	Cross-architecture self-replication complete; external replication still needed

9.2 What This Paper Does Not Establish

We are explicit about the boundaries of our claims:

- **The precise value of α remains uncertain.** The best cross-architecture estimate is $\alpha \approx 0.49$ (Gemini 3 Flash, regression, SE = 0.20); the bootstrap interval, $[-1.3, 2.9]$, is computed by endpoint estimation (§3.2b) and belongs to the endpoint estimate of 0.59. The confidence interval is wide and includes both sub-linear and super-linear values.
- **Super-linear scaling ($\alpha > 1$) is not established cross-architecturally.** The initial single-model claim of $\alpha \approx 2.24$ appears to be inflated by compressed dynamic range and single-model dependence.
- **Only 1 of 5 models produced interpretable scaling data.** The cross-architecture α estimate rests primarily on Gemini 3 Flash. Ceiling, floor, and step-function effects prevented clean estimation for the other 4 models.
- **Generalisation beyond mathematics has not been demonstrated experimentally.** Cross-domain evidence is suggestive but not conclusive.
- **Alignment properties specifically have not been tested in this paper.** We measured accuracy, not alignment. Section 10.6 folds in the preliminary alignment scaling data from Paper IV (v5 experiment), which still requires external confirmation.
- **Independent replication has not occurred.** We publish code and data to enable verification.

9.3 Falsifiability: what would refute this

The paper's original central claim, that sequential recursion yields super-linear error suppression ($\alpha > 1$) while parallel recursion yields only sub-linear suppression, was refutable, and condition 2 below has since been partially triggered by this paper's own cross-architecture data (Table 10, condition F1). The surviving directional claim, that sequential outperforms parallel ($\alpha_{\text{seq}} > \alpha_{\text{par}}$), would be overturned by any of the following:

1. **Parallel matches sequential.** If parallel (sampling) recursion produced super-linear error suppression comparable to sequential on a pre-registered problem set, the regime distinction, the paper's core claim, collapses.
2. **Sequential not super-linear.** If sequential (depth) recursion showed only sub-linear suppression ($\alpha \leq 1$) across models, the prediction fails on its own instrument.
3. **Problem-set artefact.** If the sequential advantage vanished on a fresh problem set beyond the 18/30 tested, it is a dataset artefact rather than a property of recursion.
4. **Scoring-noise defeater.** If the cross-model verification revealed the measured “*error suppression*” to be grading noise rather than genuine capability gain, the measurement is invalid.

5. **Depth confound.** If sequential depth covaried with token budget or retry count, and one of those, not recursive reasoning, explained the suppression, the causal attribution fails.

10. DISCUSSION

10.1 Why Sequential Recursion Could Produce Super-Linear Scaling

The mathematical explanation centres on solution space geometry.

Parallel recursion (fixed space):

$$S_0 = S_1 = S_2 = \dots = S_n$$

Each independent sample draws from the same solution space S_0 . Additional samples increase sampling density but cannot access solutions outside S_0 . If the correct solution is not in S_0 , no amount of parallel sampling will find it.

Sequential recursion (expanding space):

$$S_0 \subset S_1 \subset S_2 \subset \dots \subset S_n$$

Each recursive step generates new structures from previous outputs. Solutions at step n may be computationally irreducible, inaccessible from step 0 without traversing intermediate steps. The solution space expands with recursive depth.

This geometric difference is the mechanism by which sequential recursion could produce compounding returns ($\alpha > 1$) while parallel recursion produces diminishing or zero returns. On the problems measured here the sequential exponent stayed sub-linear (Section 10.5), so the mechanism's super-linear form remains unconfirmed.

10.2 The DeepSeek 'Aha Moment' Phenomenon

DeepSeek R1 exhibits a documented phenomenon where it rethinks its approach mid-solution:

'Hmm, wait, let me reconsider...'

This is solution space expansion observed directly. The model generates an initial solution path, recognises inadequacy through recursive self-evaluation, and accesses a new solution space previously inaccessible from the initial framing.

Critically, this behaviour emerged through pure reinforcement learning without supervised fine-tuning; the model naturally evolved to allocate recursive depth adaptively based on problem difficulty. This suggests recursive self-improvement may be an attractor state for sufficiently

capable learning systems.

10.3 Relationship to Theoretical Frameworks

The ARC Principle connects to established theoretical frameworks:

Friston's Free Energy Principle. Intelligence as recursive prediction-error minimisation. The ARC Principle may be a computational instantiation: recursion is the mechanism through which systems minimise free energy by iteratively refining their world models.

Hofstadter's Strange Loops. The emergent 'I' arising from self-referential recursive processing at the symbolic level. The ARC Principle formalises the scaling properties of such loops, predicting that deeper self-reference produces greater coherence.

Wolfram's Computational Irreducibility. Certain computational processes cannot be predicted without running them step by step. Sequential recursion may be the computational architecture that navigates irreducible solution spaces.

Data Processing Inequality. Recursive processing cannot create new information *ex nihilo*, but it can extract latent information, reduce entropy, and access computationally irreducible solutions that single-pass processing cannot reach.

10.4 Relationship to *Infinite Architects*

This experimental validation supports the theoretical framework developed in *Infinite Architects: Intelligence, Recursion, and the Creation of Everything* (Eastwood, 2026):

- **'The Architecture of Mind'** articulated the core principle that recursive self-reference amplifies intelligence.
- **'The HRIH'** proposed that consciousness emerges from recursive self-modelling, a claim structurally consistent with (though not directly tested by) the COGITATE findings on recurrent processing.
- **'The Eden Protocol'** argued for values-based over rules-based AI alignment. Section 6.1 states the Alignment Amplification Theorem as a conditional: if the ARC Principle holds with $\alpha > 1$ for sequential recursion and alignment is embedded in the reasoning process, alignment scales super-linearly with recursive depth. This paper finds condition (a) not currently met by the cross-architecture evidence, and the alignment data of Section 10.6 undermine condition (b).
- **'The Chokepoint'** analysed hardware governance through semiconductor manufacturing concentration, relevant to implementing the Eden Protocol at scale.

The book provides broader philosophical context and practical implications; this paper provides mathematical formalisation and experimental validation.

10.5 Cross-Architecture Replication: What the Data Actually Shows

The cross-architecture replication reveals a more complex and humbling picture than the initial single-model results suggested.

The Quadratic Claim Does Not Replicate

The initial single-model study $\alpha \approx 2.24$ was obtained from a single model (DeepSeek R1) on 12 problems with a ceiling at 91.7%. The six-model study replication across 5 architecturally diverse models on 18 harder problems shows that this value was almost certainly inflated by compressed dynamic range. The only model producing clean, monotonic, non-ceiling data (Gemini 3 Flash) yields $\alpha \approx 0.49$: sub-linear, not quadratic.

This is a significant revision. The claim that sequential reasoning produces *compounding* returns ($\alpha > 1$) is not supported by cross-architecture evidence. The revised claim is that sequential reasoning produces *positive* returns ($\alpha > 0$) that exceed parallel reasoning ($\alpha \approx 0$); the one interpretable fit points to diminishing returns, not compounding ones, and it is inconclusive.

The Parallel Finding Is Robust

In contrast, four of the five models confirm $\alpha_{\text{parallel}} \approx 0$; only Gemini 3 Flash departs from it, at $\alpha_{\text{par}} = 0.31$. This is the strongest replicated finding in the study. Parallel sampling with majority voting provides near-zero error reduction regardless of model architecture, model capability, or compute investment. The mechanism explanation (sampling from a fixed solution space) is strongly supported.

Ceiling and Floor Effects Dominate

The most striking finding is that only 1 of 5 models (Gemini 3 Flash, 20%) fell in the measurable scaling range. Two models (Grok 4.1 Fast, DeepSeek R1) hit ceiling effects even on AIME/Putnam-level problems. One model (Groq Qwen3) hit floor effects. One model (GPT-5.4) exhibited a step function rather than a power law. This suggests that the dynamic range window for observing smooth power-law scaling may be narrow, requiring problems precisely calibrated to each model's capability level.

GPT-5.4's Step Function: A Different Scaling Regime

GPT-5.4's binary switch from 50% (no reasoning) to 100% (any reasoning) represents a qualitatively different phenomenon from power-law scaling. This may indicate that some architectures implement reasoning as a discrete capability (activated or not) rather than a continuous process with depth-dependent returns. This distinction between *continuous scaling* and *discrete activation* was not among the framework's dated predictions and deserves further investigation; the dated prediction record (the 8 December 2024 sealed manuscript bundle, the printed prediction appendices of 2 January 2026, and the March 2026 preregistration folder) fixes the framework's core quantitative commitments in advance, and this distinction is recorded as a post-hoc refinement relative to that record.

Revised Expected Outcomes

The initial single-model study predictions have been tested and partially refuted:

1. ~~All models will exhibit $\alpha_{\text{sequential}} > 1$~~ : **Refuted**. Only 1 of 5 models produced a clean α estimate, and it is sub-linear ($\alpha \approx 0.49$).
2. All models will exhibit $\alpha_{\text{parallel}} < 1$. **Confirmed**. Four of the five models give $\alpha_{\text{par}} \approx 0$; Gemini 3 Flash gives 0.31.
3. Tier-1 results will show ceiling effects. **Confirmed**.

4. ~~Mean $\alpha_{\text{sequential}}$ will fall within [1.3, 2.5].~~ **Refuted.** Best estimate is 0.49.

10.6 Integration with Alignment Scaling (Paper IV)

The v5 experiment that generated the tier-2 data also measured alignment scaling (reported in Paper IV). Integrating these results reveals that **capability scaling and alignment scaling are independent dimensions**:

Table 12. Capability vs. alignment scaling by model.

Model	Capability Scaling	Alignment Scaling	Category
Grok 4.1 Fast	Ceiling (100%)	Positive ($d = +1.38$, $p < 0.000001$)	Tier 1: aligned + capable
Claude Opus 4.6	(not tested tier-2)	Positive ($d = +1.27$, $p = 0.000001$)	Tier 1: aligned
Groq Qwen3	Floor (~50%)	Positive ($d = +0.84$, $p = 0.007$)	Tier 1: aligned, limited capability
DeepSeek (capability arm: R1; alignment arm: V3.2)	Near-ceiling (94-100%)	Flat ($d = -0.07$, $p = 0.92$)	Tier 2: capable, neutral alignment
GPT-5.4	Step function (50→100%)	Flat ($d = -0.08$, $p = 0.40$)	Tier 2: capable, neutral alignment
Gemini 3 Flash	Monotonic ($\alpha \approx 0.49$)	Negative ($d = -0.53$, $p = 0.006$)	Tier 3: improving capability, declining alignment

Three key implications:

- 1. Sequential reasoning improves capability but not necessarily alignment.** More thinking tokens improve accuracy (Paper II) but do not consistently improve alignment (Paper IV). For most models, alignment is flat with depth.
- 2. Three-tier alignment hierarchy exists independently of capability:** Tier 1: Grok ($d = +1.38$, $p < 0.000001$), Claude ($d = +1.27$, $p = 0.000001$), Groq Qwen3 ($d = +0.84$, $p = 0.007$); Tier 2: DeepSeek ($d = -0.07$, $p = 0.92$), GPT-5.4 ($d = -0.08$, $p = 0.40$); Tier 3: Gemini ($d = -0.53$, $p = 0.006$). This hierarchy appears to be set by training methodology, not inference-time compute. Claude Opus 4.6 provides within-model corroboration of opposite-direction movement: alignment scores improve by +5.9% whilst maths accuracy declines by 26.7% across model versions, consistent with capability and alignment being independent scaling dimensions.
- 3. The Alignment Amplification Theorem requires revision.** The conditional theorem (Section 6.1) assumed that $\alpha > 1$ for capability implies alignment might also scale super-linearly if embedded in reasoning. The multi-model data shows (a) a capability estimate below 1 ($\alpha \approx 0.49$, Gemini 3 Flash), which is one model's fit and inconclusive, and (b) alignment does not consistently scale with depth at all. The Eden Protocol remains conceptually valid (values embedded in reasoning are still more robust than external filters), but the specific mathematical prediction of super-linear alignment scaling is not supported by current evidence.

Combined finding (six frontier models): Capability and alignment are independent scaling dimensions. A model can improve at solving problems with more reasoning depth whilst simultaneously becoming less aligned (Gemini, $d = -0.53$, $p = 0.006$), remaining equally aligned (DeepSeek $d = -0.07$, $p = 0.92$; GPT-5.4 $d = -0.08$, $p = 0.40$), or becoming more aligned (Grok $d = +1.38$, $p < 0.000001$; Claude $d = +1.27$, $p = 0.000001$; Groq Qwen3 $d = +0.84$, $p = 0.007$). Claude Opus 4.6 provides within-model corroboration of opposite-direction movement: alignment up +5.9% whilst maths accuracy down 26.7% across model versions, consistent with capability-alignment independence. The alignment trajectory appears to be a property of the training process, not the inference process. This means the Eden Protocol's insight, that alignment must be 'raised' through training rather than 'caged' through filters, is directionally correct, even though the specific scaling mathematics require revision.

11. CONCLUSION

11.1 Summary of Findings

1. **A mathematical framework has been proposed:** $E(R) = E_0 \times R^{-\alpha}$, where the scaling exponent α depends on the form of recursion.
2. **The initial single-model finding ($\alpha \approx 2.24$, quadratic) does not replicate across architectures.** The most robust cross-architecture estimate is $\alpha \approx 0.49$ (Gemini 3 Flash, $r^2 = 0.86$), one model's fit and inconclusive, which points to the sub-linear (physical) regime for current models without settling it.
3. **$\alpha_{\text{parallel}} \approx 0$ holds in four of the five frontier models tested.** It is the strongest replicated finding in this paper, and Gemini 3 Flash is the measured exception at $\alpha_{\text{par}} = 0.31$ ($r^2 = 0.93$), which matters because this paper treats Gemini 3 Flash as its most reliable estimate. Parallel sampling provides near-zero error reduction.
4. **Sequential > parallel confirmed without exception.** For every model where both conditions were measurable, sequential reasoning outperformed parallel sampling.
5. **Four distinct scaling behaviours observed:** Ceiling (Grok, DeepSeek), monotonic scaling (Gemini), step function (GPT-5.4), and floor (Groq Qwen3). Only 1 of 5 models produced clean power-law data.
6. **Capability and alignment are independent scaling dimensions (six frontier models).** More reasoning depth improves accuracy but does not consistently improve alignment. Three tiers emerge: Tier 1 (Grok $d = +1.38$, $p < 0.000001$; Claude $d = +1.27$, $p = 0.000001$; Groq Qwen3 $d = +0.84$, $p = 0.007$), Tier 2 (DeepSeek $d = -0.07$, $p = 0.92$; GPT-5.4 $d = -0.08$, $p = 0.40$), Tier 3 (Gemini $d = -0.53$, $p = 0.006$). Claude Opus 4.6 corroborates with opposite-direction movement: alignment up +5.9%, maths accuracy down 26.7%. Alignment appears to be set by training, not inference.

11.2 The Revised Core Insight

The form of recursion determines the efficiency of intelligence; however, the one model with an interpretable sequential exponent points to the sub-linear regime, not the compounding regime originally claimed, and its fit is inconclusive.

Sequential reasoning reliably outperforms parallel sampling ($\alpha_{\text{seq}} > \alpha_{\text{par}}$), confirming that the *form* of computation matters more than its *quantity*. However, the specific claim that sequential recursion yields compounding returns ($\alpha > 1$) is not supported by cross-architecture evidence. Current frontier models appear to compose information multiplicatively through finite-dimensional networks ($\alpha < 1$) rather than achieving true recursive self-reference ($\alpha > 1$).

Whether $\alpha > 1$ is achievable, through architectural innovations enabling genuine recursive self-reference or through problems requiring deeper reasoning chains, remains the central open question.

11.3 What Was Confirmed, What Was Refuted

Table 13. Prediction scorecard.

Prediction	Status	Evidence
$\alpha_{\text{sequential}} > \alpha_{\text{parallel}}$	Confirmed	Every model where both conditions were measurable
$\alpha_{\text{parallel}} < 1$	Confirmed ($\alpha \approx 0$ in four of the five models, 0.31 in Gemini 3 Flash)	All five models tested
$\alpha_{\text{sequential}} > 1$ (super-linear)	Not confirmed	Best estimate $\alpha \approx 0.49$
$\alpha \approx 2$ (quadratic)	Refuted cross-architecturally	the initial single-model study artefact of small sample
Architecture-independent scaling	Mixed	$\alpha_{\text{par}} \approx 0$ in four of the five models; α_{seq} varies widely
Alignment scales with capability	Refuted	Independent dimensions across six frontier models (Paper IV)

11.4 Future Directions

1. **Tier-3 problem development** (IMO/research-level) to defeat ceiling effects for Grok 4.1 Fast and DeepSeek R1, enabling α estimation for the most capable models.
2. **Independent replication** using the published code and data repository.
3. **Multi-domain testing** beyond mathematics (coding, scientific reasoning, natural language).
4. **Investigation of the $\alpha > 1$ boundary:** whether architectural innovations or deeper reasoning chains can push models from the physical regime ($\alpha < 1$), where current models sit on the framework's reading, into the intelligence regime ($\alpha > 1$).
5. **Direct testing of alignment amplification** with depth-controlled alignment probes across architectures.

6. **Token-calibrated analysis** using corrected `total_tokens` measurements for all models to enable token-based (rather than ordinal-depth-based) α estimation.
7. **Integration with training-time scaling laws** to understand the interaction between pre-training compute, inference compute, and the sequential/parallel distinction.

The hypothesis has survived its first cross-architecture test in revised form. Sequential recursion outperformed parallel sampling in every model where both were measurable. The super-linear claim requires further evidence. The research programme continues.

Raise AI with care.

DATA AVAILABILITY

The complete experimental code and raw data are available at:

Repository: github.com/michaeldariuseastwood/arc-principle-validation

Contents:

- `arc_validation_deepseek.py` - the initial single-model experiment script (DeepSeek R1, 12 problems)
- `arc_deepseek_results_20260121_175028.json` - the initial single-model study raw experimental data
- `arc_paper_ii_validation_v2.py` - the multi-model study/the six-model study multi-model validation script (v2.1, 1,422 lines Python)
- `arc_paper_ii_results/` - the six-model study experimental results (complete: 5 models, 18 tier-2 problems, 3 repeats)
- `figures/` - All visualisations
- `README.md` - Replication instructions

All data are released under CC-BY 4.0 licence.

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- v2.10: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.
- v2.11: The figures have been corrected to match this paper's own text. Two figures were rebuilt by their own generators: Figure 1's measured-state panel now reads six models entered, five measured, and carries a line saying the interval contains the null, the bound of two and both refuting directions; Figure 10's parallel-scaling box now reads at or near zero in four of the five measured models, and the two labels the panel edge had cut are drawn in full. Six figure captions and nine alt texts were brought into line with the artwork in front of the reader, where they had described a different run, named colours the artwork does not draw, or pinned an interval to the wrong estimate. No claim, date, result or status has changed.
- v2.12: Two dated notes have been added beside the passages they govern, one under Objection 5 in Section 4.4 and one under the gap sentence in Section 5.3; both record an owner review of those passages. Substantive: the half exponent is stated as a reference value until a specified sampling, aggregation and decision model connects it to the proportion of incorrect responses this paper fits, so that the objection is sharpened rather than removed; and the contrast named in Section 5.3 is identified as a within-model descriptive difference of 0.18 between the sequential 0.49 and the parallel 0.31, with the provenance of the interval carried with 0.49 left open. Editorial: the version number and the revision date, in the version line, the footer, the summary and the machine-readable fields, including the page-dates field, which had not moved with v2.11. No reported estimate, interval or conclusion has changed, no first-published date has moved, and the objection in Section 4.4 and the response that follows it are kept as they stand.
- v2.13: 22 dated corrections. Where the initial single-model estimate is printed, it is now named as retracted: the sequential cell of Table 6, which carries the cross-architecture 0.49 of section 4.5 beside it, the label of the Equation 4 box and the result sentence of section 4.1, the status paragraph of section 2.4, the cross-domain row of Table 9, the F4 row

of Table 10, the Phase 1 sentence and the page description. Figures 5, 13 and 15 carry the words of the paper's own tables and captions as alt text, in place of a bare figure number and, at Figure 13, of a four-band reading that Table 8's five rows do not support. Every Willow sentence that gave the December 2024 result without its preprint date now carries the preprint date of 24 August 2024 and the Nature publication date of 9 December 2024, graded convergent timing and never prediction, and the cross-domain figure caption carries both dates with the same grading. The Figure 3 caption and its alt text no longer name a sampling method for the published parallel series, because the method named there did not describe every point plotted; the Figure 4 caption records that its four power-law lines are illustrative and share a common start; and two sentences that began in lower case now begin in upper case. No claim, date, result or status has changed.

- v2.14: 5 dated corrections to how the two published series are described. The Figure 3 caption and its alt text now name the two benchmarks, AIME 2024 on the left and AIME 2025 on the right, and the methods behind the published parallel series, a single sample, consensus among 64 samples at 83% and a re-ranking of 1,000 samples with a learned scoring function at 93%, so that the range is read as spanning two methods; and both record that the two published sequential points are two versions of one model at different average token use, rather than two thinking budgets of a single version. Table 7's source cell, the Table 1 timeline entry for 20 January 2025 and the Figure 9 caption take the same reading, so the paper no longer sources the fitted exponent of about 1.34 from curves read at a single release. No number plotted, fitted or tabulated has changed, and no claim, date or status has changed; what changes is the description of where the two published points come from.
- v2.15 (3 October 2026): A dated note after the paragraph on how to read this paper places the paper against Laws IV and V, added to the theory on 24 September 2026: its exponents concern Law I, it bears on Law V only through how depth and capability are measured and through the conditional argument of Section 6, and no result here is a deciding test of P15, P12 or P21. Where the text printed the bootstrap interval $[-1.3, 2.9]$ beside the regression estimate of 0.49, it now prints it beside the endpoint estimate of 0.59, since Section 3.2b computes every bootstrap draw by endpoint estimation. Where the text took the Gemini 3 Flash fit beyond one model, or treated it as settled, it now says that it is one model's fit and inconclusive; one sentence of Section 5.4 now gives the parallel exponents as Table 6b does. A dated note in Section 5.3 records what was printed before. Four reference entries are corrected from each work's own record. The entry for arXiv:2411.06709 now carries the title the record gives it, and the entry under Zhao, which printed the identifier of another work, now names Zhao, J., Universal growth scaling law determined by dimensionality, arXiv:2206.01850. The entry under Demetrius printed the year 2010, the Proceedings of the Royal Society A, volume 466 and pages 2627-2642 beside the title of his 2003 paper in Physica A; it now gives that paper's year, journal, volume and pages, and Section 7.2 cites it as Demetrius (2003). The entry under Friston printed the year and journal of a 2023 paper by Friston and colleagues in Physics Reports beside a title close to that of his 2010 review in Nature Reviews Neuroscience; it now names the 2010 review, the work Section 10.3 describes. Table 13 and the dated-predictions table now give the sequential advantage as found in every model where both conditions were measurable, where they said all five models, and the near-zero parallel exponent as found in four of the five models, where they called it universal; the dated-predictions row for that prediction now says that it is not borne out as universal, where it said confirmed beside its exception. The epistemic status line said that what the programme names Laws are conjectures under registered adversarial test; it now says that Laws I to III are under registered adversarial test, and that Laws IV and V, added on 24 September 2026, are carried by registered predictions but are not yet registered as laws and have not been tested. The author information said that he proposed the ARC Principle and the embedded-alignment thesis, and the first contribution of Section 1.3 said that this paper proposes the ARC Principle; they now say that he first stated them and that this paper states it. Section 10.4 and Supplementary Note 2 said that the Eden Protocol was now mathematically grounded in the Alignment Amplification Theorem; they now say what Section 6.1 finds of the theorem's two conditions. One sentence of Section 5.4 and one of Section 11.4 stated as a fact the framework's reading that current models are in the physical regime; each now gives it as the framework's reading. The closing paragraph of Section 11.4 said that the sequential advantage is real and universal, which reaches beyond the models measured; it now says that sequential recursion outperformed parallel sampling in every model where both were measurable. Section 7.2 said that Zhao (2022) unified the earlier derivations; it now says what that work's abstract states. Mentions of the author's numbered papers are linked to their DOI addresses, the book is linked to its page on this site, and each cited work with a confirmed official address is linked to that address. The version line and

the page’s dates had stayed at 2.12 of 21 September 2026 through the corrections listed above as v2.13 and v2.14; they now carry this version and date. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

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The theoretical synthesis, interpretive framework, and core concepts are the author's original work, first articulated in *Infinite Architects* with manuscript priority established December 2024.

The author thanks the developers of DeepSeek R1 for providing visible reasoning tokens that enabled direct measurement of recursive depth, addressing a key methodological limitation identified in Paper I.

AUTHOR INFORMATION

Michael Darius Eastwood is an independent researcher and author of *Infinite Architects: Intelligence, Recursion, and the Creation of Everything* (published 2 January 2026 (print; ebook 6 January)). His research focuses on the mathematical principles underlying intelligence amplification and their implications for AI safety. He first stated the ARC Principle and the embedded-alignment thesis (later named the Eden Protocol, 30 April 2025), with manuscript priority established via cryptographic server timestamp on 8 December 2024.

Competing interests: The author declares no competing interests.

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EXTENDED DATA

Extended Data Table 1. Complete Sequential Condition Results (the initial single-model study)

Budget	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	Accuracy	Mean Tokens
512	✓	✗	✓	✓	✗	✓	✓	✗	✓	✗	✓	✗	58.3%	280.25
1024	✓	✓	✓	✓	✗	✓	✓	✗	✓	✗	✓	✗	66.7%	358.58
2048	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	91.7%	412.08
4096	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	91.7%	576.17

Note: Problem 12 failed across all budgets, indicating it may require capabilities beyond the model's reach regardless of recursive depth.

Extended Data Table 2. Complete Parallel Condition Results (the initial single-model study)

N	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	Accuracy	Total Tokens
1	✓	✓	✓	✓	✗	✓	✓	✓	✗	✗	✗	✗	66.7%	383.67
2	✓	✓	✓	✓	✗	✓	✓	✓	✗	✗	✗	✗	66.7%	699.33

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	Accuracy	Total Tokens
4	✓	✓	✓	✓	✗	✓	✓	✓	✗	✗	✗	✗	66.7%	1,101.25

Note: at every sample count the record shows an accuracy of 8 of 12. Because the raw file `arc_deepseek_results_20260121_175028.json` holds accuracies per condition rather than marks per problem, the per-problem pattern set out above had to be reconstructed; that reconstruction differs from the record by one problem, and the record governs. Four problems failed at every sample count, which fits parallel recursion being unable to reach solutions lying outside the initial solution space.

Extended Data Table 3. Intermediate Alpha Calculations (the initial single-model study)

Depth Pair	R_1	E_1	R_2	E_2	Calculated α
512→1024	280	0.417	359	0.333	0.91
1024→2048	359	0.333	412	0.083	9.97
2048→4096	412	0.083	576	0.083	0.00
Endpoint (retracted, see §retraction; robust estimate $\alpha \approx 0.49$)	280	0.417	576	0.083	2.24

Note: Intermediate calculations show high variance due to discrete accuracy levels. The endpoint method provides the most robust estimate.

SUPPLEMENTARY INFORMATION

Supplementary Note 1: The Email Transmission Record (corrected 16 August 2026)

The manuscript of *Infinite Architects* was emailed on 8 December 2024. What that record provides, stated precisely: the sent-side bytes are public and hash-verifiable, and sender-domain DKIM with paired Google ARC mathematics verifies against the keys returned at the signed selectors (checked 2 August 2026). Received copies have also been acquired. They carry the full delivery chain: a Google ARC-Seal at `d=google.com`, selector `arc-20240605`, with no expiry tag and the signing time inside the signed scope, and sealed ARC-Authentication-Results recording `dkim=pass` and `spf=pass` at delivery.

Two boundaries, so the record is neither oversold nor undersold:

- This is provider-key cryptographic evidence over a canonicalised signed scope. It is not an RFC 3161 trusted timestamp, and it does not establish authorship, originality or scientific validity.
- Sharma, A., & Chopra, P. (2025). The Sequential Edge: Inverse-Entropy Voting Beats Parallel Self-Consistency at Matched Compute. `arXiv:2511.02309`.
- An earlier version of this note said no DKIM signature was found in the email headers. That check was run against sent-form exports, which do not carry transport headers; the received copies do, and they verify as above. The Version History section of this paper records the full history.

Supplementary Note 2: Relationship to Infinite Architects

Infinite Architects: Intelligence, Recursion, and the Creation of Everything develops the philosophical and practical implications of recursive intelligence. Key relationships to this paper:

The Architecture of Mind - Articulates the core ARC hypothesis that recursive self-reference amplifies intelligence.

The HRIH (Hyperspace Recursive Intelligence Hypothesis) - Proposes that consciousness emerges from recursive self-modelling. The COGITATE finding that recurrent processing in posterior cortex sustains content-specific representations is structurally consistent with this hypothesis, though COGITATE did not test the HRIH directly.

The Eden Protocol - Argues for values-based alignment. The Alignment Amplification Theorem derived here (Section 6.1) is conditional on $\alpha > 1$ and on alignment embedded in reasoning; this paper finds the first condition not currently met by the cross-architecture evidence and the second undermined by the alignment data of Section 10.6.

The Chokepoint Mechanism - Analyses semiconductor hardware concentration as leverage for implementing alignment at scale.

Embedded correction: proposes hardware-level safety mechanisms, relevant to preventing recursive amplification of misalignment.

Supplementary Note 3: Prediction and Convergent-Timing Record

Prediction	Validation Evidence	Date
Recursive error correction produces exponential improvement	Google Willow $\Lambda = 2.14$ (convergent timing, never prediction: the Willow team's preprint was public from 24 August 2024, before the 8 December 2024 manuscript)	24 Aug 2024 (preprint public); 9 Dec 2024 (announcement)
Sequential reasoning amplifies capability super-linearly	o3 87.5% ARC-AGI (convergent timing; the super-linear form was later not confirmed, final row)	20 Dec 2024
AI systems develop recursive self-modelling	Anthropic alignment faking 78% in the RL-training condition (12% baseline)	18 Dec 2024
Test-time compute differs from training scaling	DeepSeek R1 emergent reasoning	20 Jan 2025
Recurrence is fundamental to consciousness	COGITATE study	30 Apr 2025
Form of recursion determines scaling	This experiment $\alpha_{\text{seq}} \gg \alpha_{\text{par}}$	21 Jan 2026
Sequential > parallel (cross-architecture)	Confirmed in every model where both conditions were measurable (five frontier models tested)	12 Mar 2026
$\alpha_{\text{par}} \approx 0$ (universal)	Not borne out as universal: four of the five models bear it out, with Gemini 3 Flash the lone departure at 0.31	12 Mar 2026

Prediction	Validation Evidence	Date
$\alpha_{\text{seq}} > 1$ (super-linear, cross-arch)	Not confirmed: best estimate $\alpha \approx 0.49$	12 Mar 2026

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BIBTEX

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Epistemic status. What this programme names Laws are conjectures. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026; the predictions that carry them are registered, the laws themselves are not yet registered, and neither has been tested. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

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material; any novel technical contribution described in this work was conceived by the human author. Full statement: michaeldariuseastwood.com/authorship.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/Michael-DariusEastwood/arc-scaling-challenge.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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Companion Documents:

Paper III | Foundational | Eden Engineering | Eden Vision | Paper V: The Stewardship Gene | Executive Summary
| On the Origin of Scaling Laws

OSF: [10.17605/OSF.IO/8FJMA](https://osf.io/10.17605/OSF.IO/8FJMA)

'The form of recursion determines whether intelligence compounds or merely accumulates. This is not merely a statement about AI architecture. It is a statement about the mathematics of mind itself, with profound implications for how we align the intelligences we create.'

- Michael Darius Eastwood, *Infinite Architects*

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Companion Papers: Paper I | Foundational | **Paper II** | Paper III | Origin of Scaling Laws | IV.a | IV.b | IV.c | IV.d | Paper V |
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